

# AI & Deep Learning

ACTL3143 & ACTL5111 Deep Learning for Actuaries  
Patrick Laub

# Lecture Outline

- **Artificial Intelligence**
- Deep Learning Successes
- Types of Machine Learning Tasks
- Neural Networks
- California House Price Prediction
- Our First Neural Network
- Force positive predictions
- Preprocessing
- Early Stopping

# Different goals of AI

Artificial intelligence describes an agent which is capable of:

|                  |                     |
|------------------|---------------------|
| Thinking humanly | Thinking rationally |
| Acting humanly   | Acting rationally   |

Put another way, what fields are most important to AI?

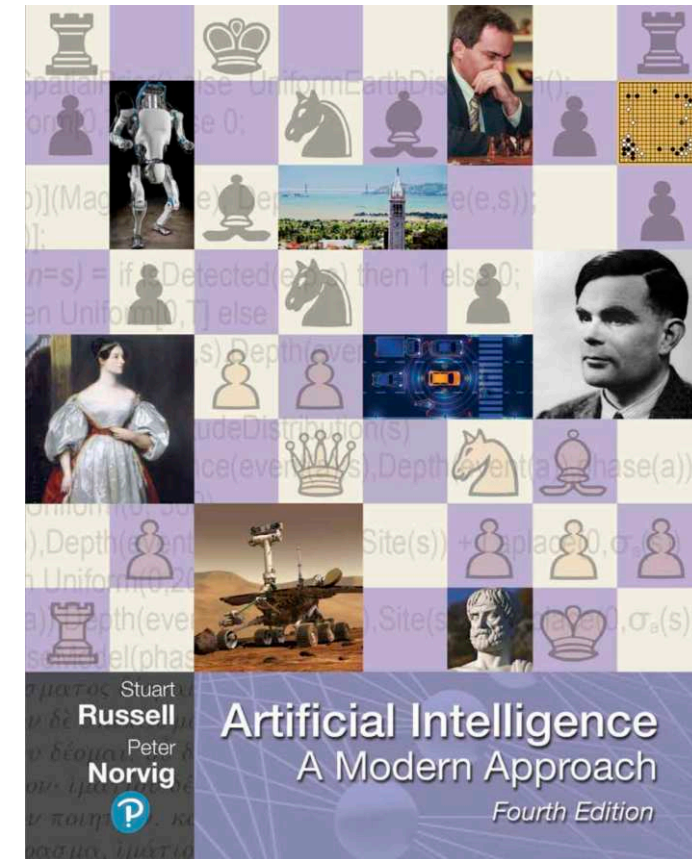
|                               |                                  |
|-------------------------------|----------------------------------|
| Cognitive science, psychology | Mathematics, logic and inference |
| HCI, linguistics and robotics | Computer science, statistics     |

- Actions are simpler to work with than thought: How do humans even think?
- Acting humanly can be done without intelligence, see ChatGPT
- The focus on actions: delivers results, but in a black-box manner

# The rational behaviour paradigm won out

“For these reasons, the rational-agent approach to AI has prevailed throughout most of the field’s history. In the early decades, rational agents were built on logical foundations and formed definite plans to achieve specific goals. Later, methods based on probability theory and machine learning allowed the creation of agents that could make decisions under uncertainty to attain the best expected outcome. In a nutshell, *AI has focused on the study and construction of agents that **do the right thing.***”

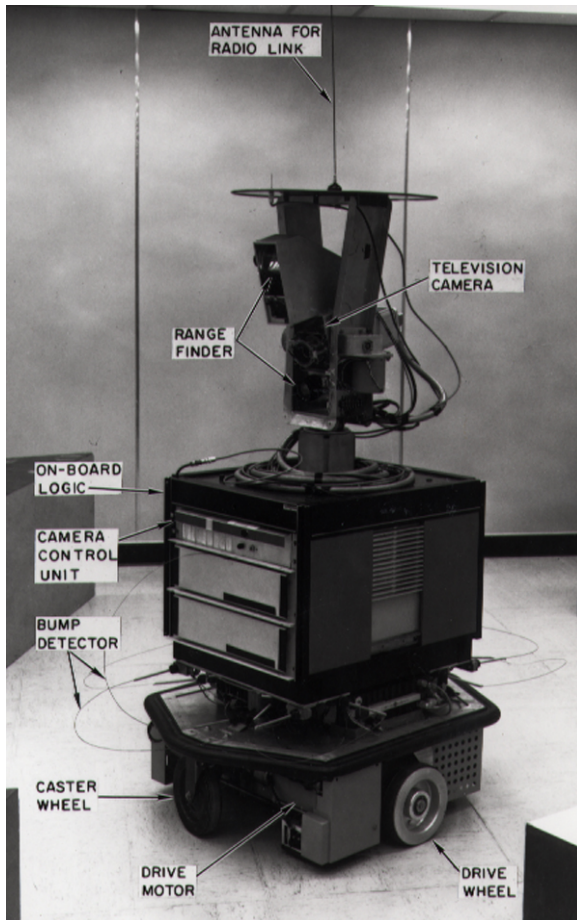
— Russell & Norvig (2021, p. 22)



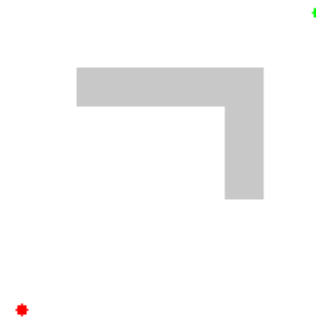
The traditional AI text.

**Question:** When do you think the term “machine learning” was first coined?

# Shakey the Robot (~1966 – 1972)

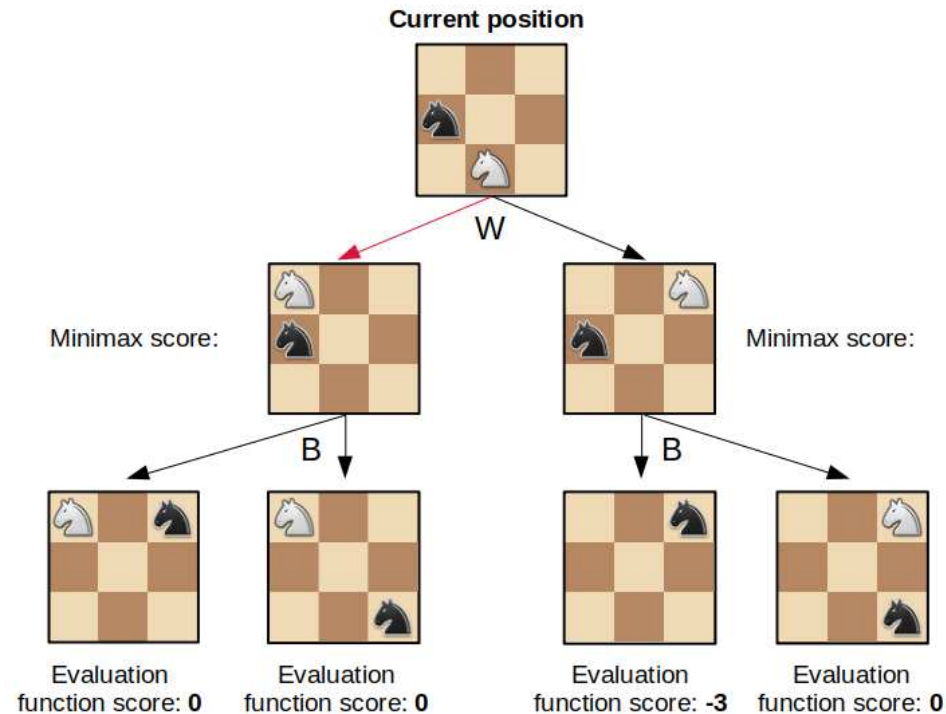


Shakey the Robot



Source: Wikipedia page for the [Shakey Project](#) and for the [A\\* search algorithm](#).

# The minimax algorithm



```
function minimax(position, depth, maximizingPlayer)
  if depth == 0 or game over in position
    return static evaluation of position

  if maximizingPlayer
    maxEval = -infinity
    for each child of position
      eval = minimax(child, depth - 1, false)
      maxEval = max(maxEval, eval)
    return maxEval

  else
    minEval = +infinity
    for each child of position
      eval = minimax(child, depth - 1, true)
      minEval = min(minEval, eval)
    return minEval
```

Pseudocode for the minimax algorithm.

The minimax algorithm for chess.



# Chess

Deep Blue (1997)



Gary Kasparov playing Deep Blue.



Cartoon of the match.

# Machine Learning

Tried *making a computer smart*, too hard!

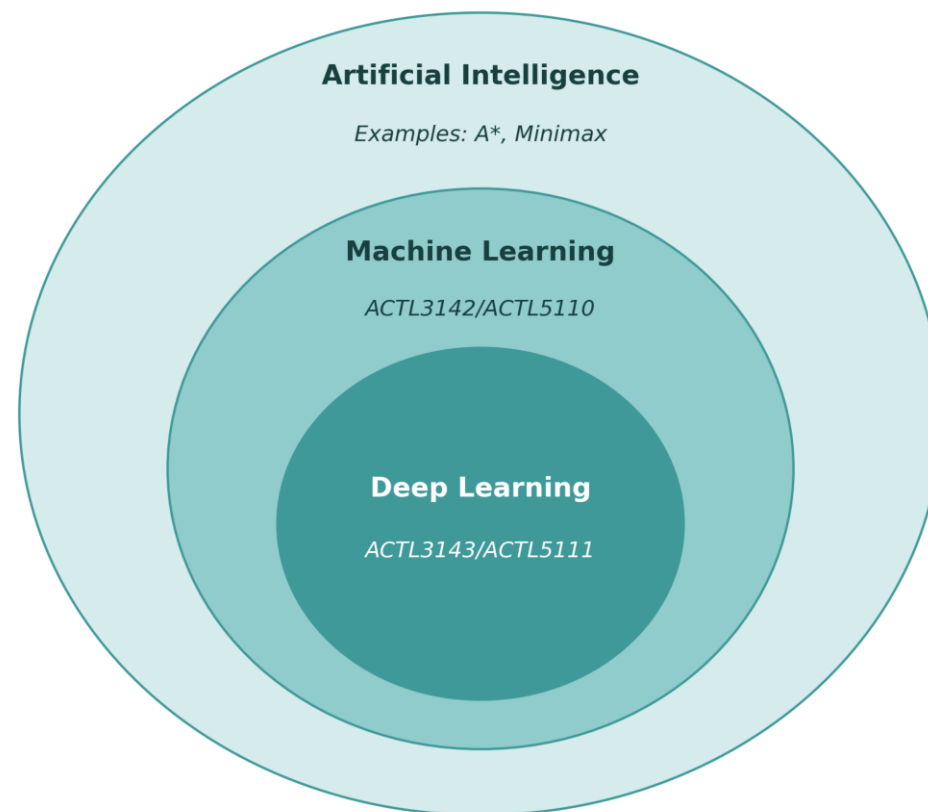
Make a computer that can **learn** to be smart.

“[Machine Learning is the] field of study that gives computers the ability to learn without being explicitly programmed”

— Samuel (1959)

AI eventually become dominated by one approach, called *machine learning*, which itself is now dominated by *deep learning* (neural networks).

You can study a 12 week course on AI and never touch on machine learning...



Artificial Intelligence, Machine Learning, and Deep Learning.



# Lecture Outline

- Artificial Intelligence
- **Deep Learning Successes**
- Types of Machine Learning Tasks
- Neural Networks
- California House Price Prediction
- Our First Neural Network
- Force positive predictions
- Preprocessing
- Early Stopping

# Image Classification I

What is this?



Options:

1. punching bag
2. goblet
3. red wine
4. hourglass
5. balloon

Source: [Wikipedia](#)

# Image Classification II

What is this?



Options:

1. sea urchin
2. porcupine
3. echidna
4. platypus
5. quill

Source: [Wikipedia](#)

# Image Classification III

What is this?



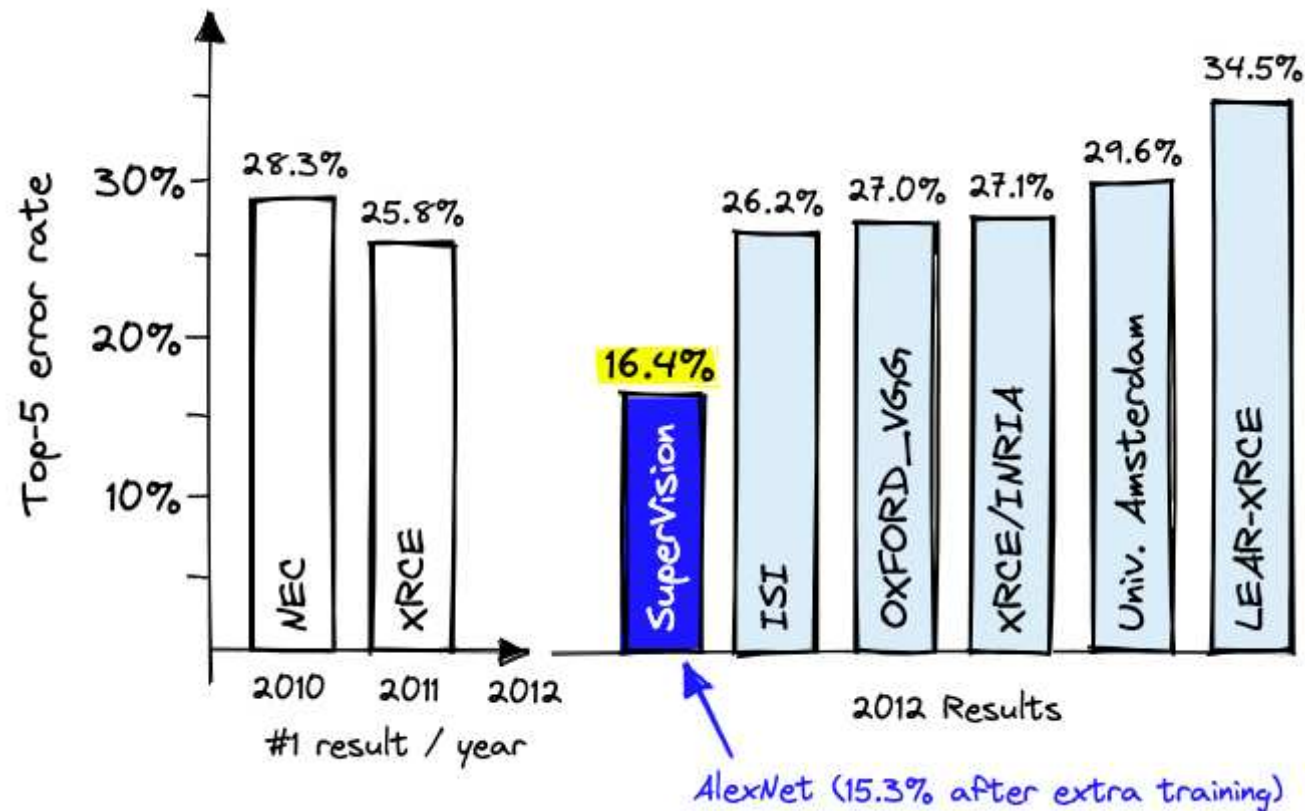
Options:

1. dingo
2. malinois
3. German shepherd
4. muzzle
5. kelpie

Source: [Wikipedia](#)

# ImageNet Challenge (2012)

ImageNet and the *ImageNet Large Scale Visual Recognition Challenge (ILSVRC)*; originally 1,000 synsets.



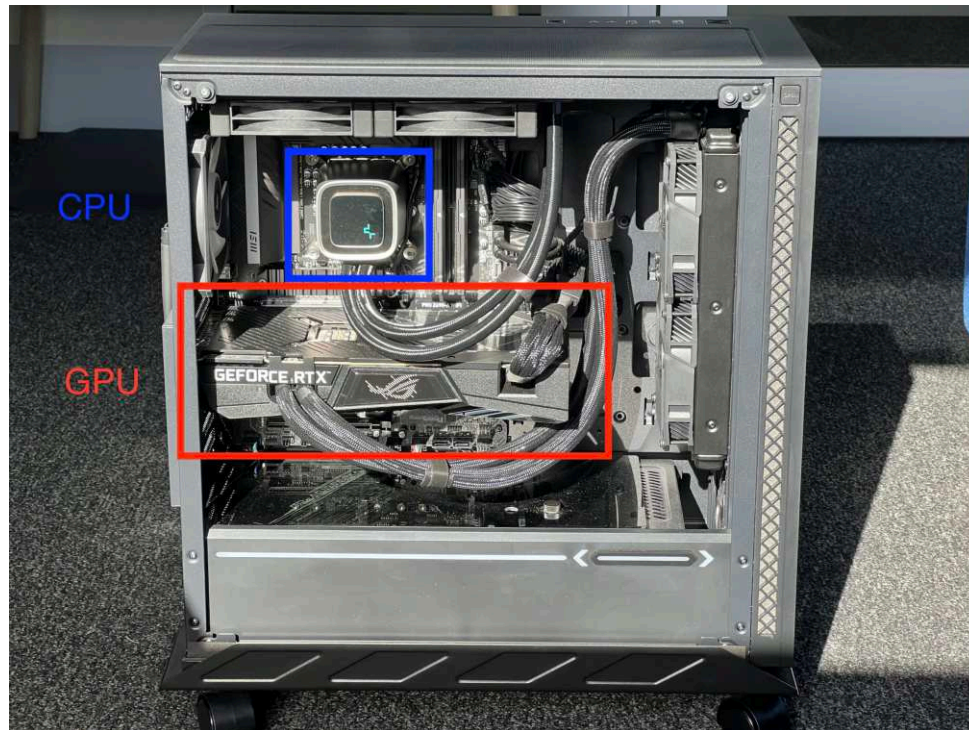
AlexNet — a neural network developed by Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton — won the ILSVRC 2012 challenge convincingly.

Source: James Briggs & Laura Carnevali, *AlexNet and ImageNet: The Birth of Deep Learning*, Embedding Methods for Image Search, Pinecone Blog



# Needed a graphics card

A graphics processing unit (GPU)



A PC with the GPU and CPU marked in red and blue.

**“4.2. Training on multiple GPUs** A single GTX 580 GPU has only 3GB of memory, which limits the maximum size of the networks that can be trained on it. It turns out that 1.2 million training examples are enough to train networks which are too big to fit on one GPU. Therefore we spread the net across two GPUs.”

— Krizhevsky et al. (2012)

# Lee Sedol plays AlphaGo (2016)

Deep Blue was a win for AI, AlphaGo a win for DL.



Lee Sedol playing AlphaGo AI

I highly recommend [this documentary about the event](#).

Source: Patrick House (2016), [AlphaGo, Lee Sedol, and the Reassuring Future of Humans and Machines](#), New Yorker article.

# Generative Adversarial Networks (2014)

<https://thispersondoesnotexist.com/>



A GAN-generated face



A GAN-generated face

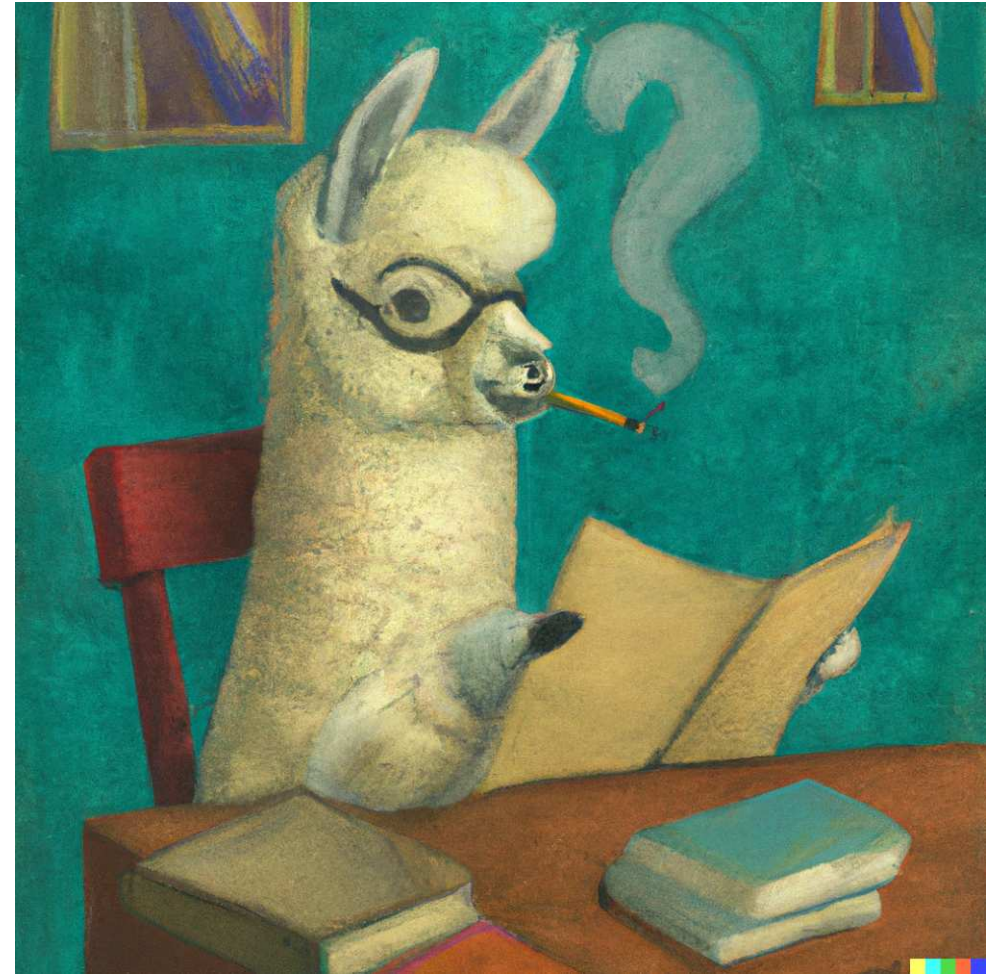
Source: Goodfellow et al. (2014).



# Diffusion models (2020)



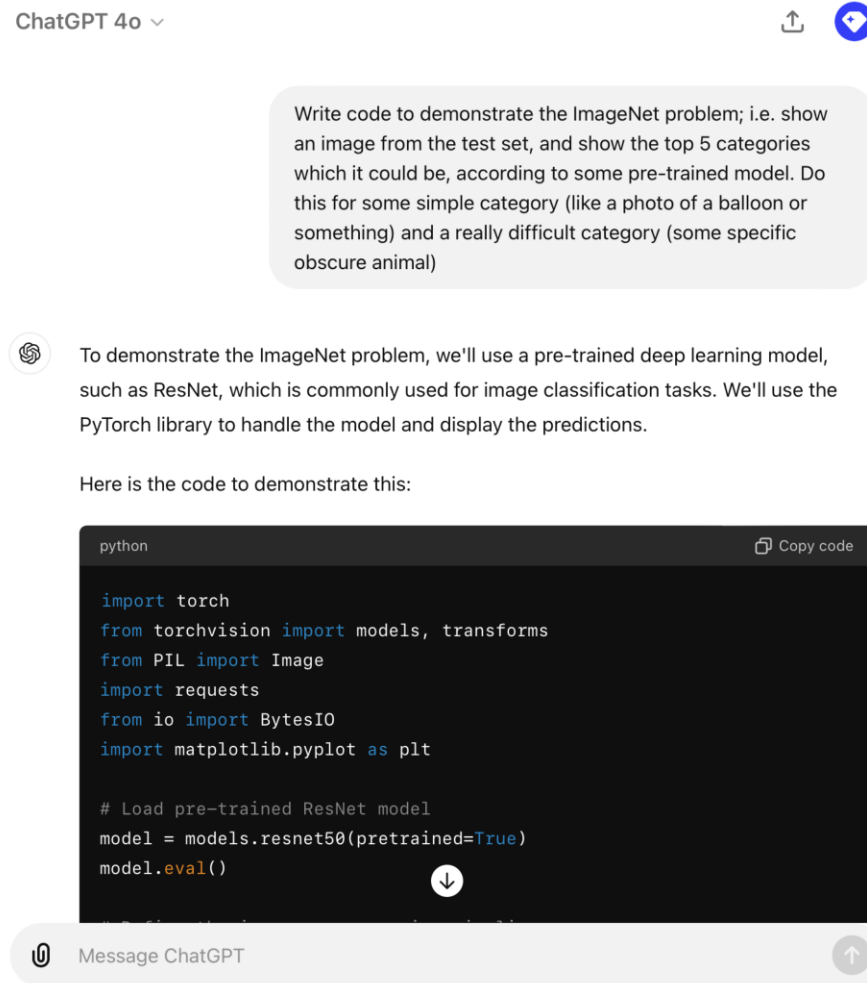
Painting of avocado skating while wearing a hoodie



A surrealist painting of an alpaca studying for an exam

Source: Ho et al. (2020). Images generated with Dall-E 2, prompts by ACTL3143 students in 2022.

# ChatGPT (2022)



Test ChatGPT's ability to:

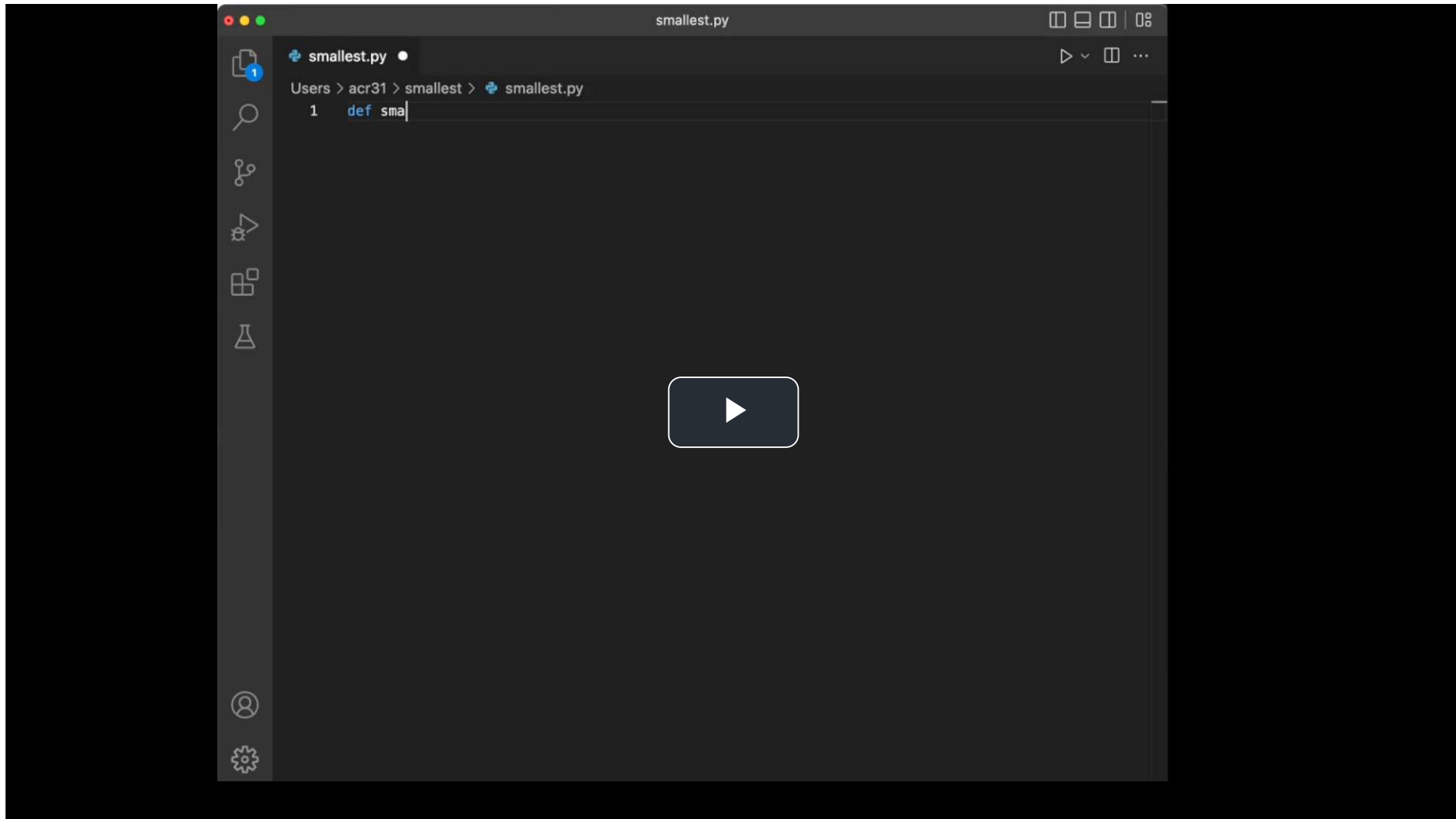
- generate images
- translate code
- explain code
- run code
- analyse a dataset
- critique code
- critique writing
- voice chat with you

Compare to Copilot.

AI predictions in the ImageNet demo were from ChatGPT code.



# GitHub Copilot (2022)

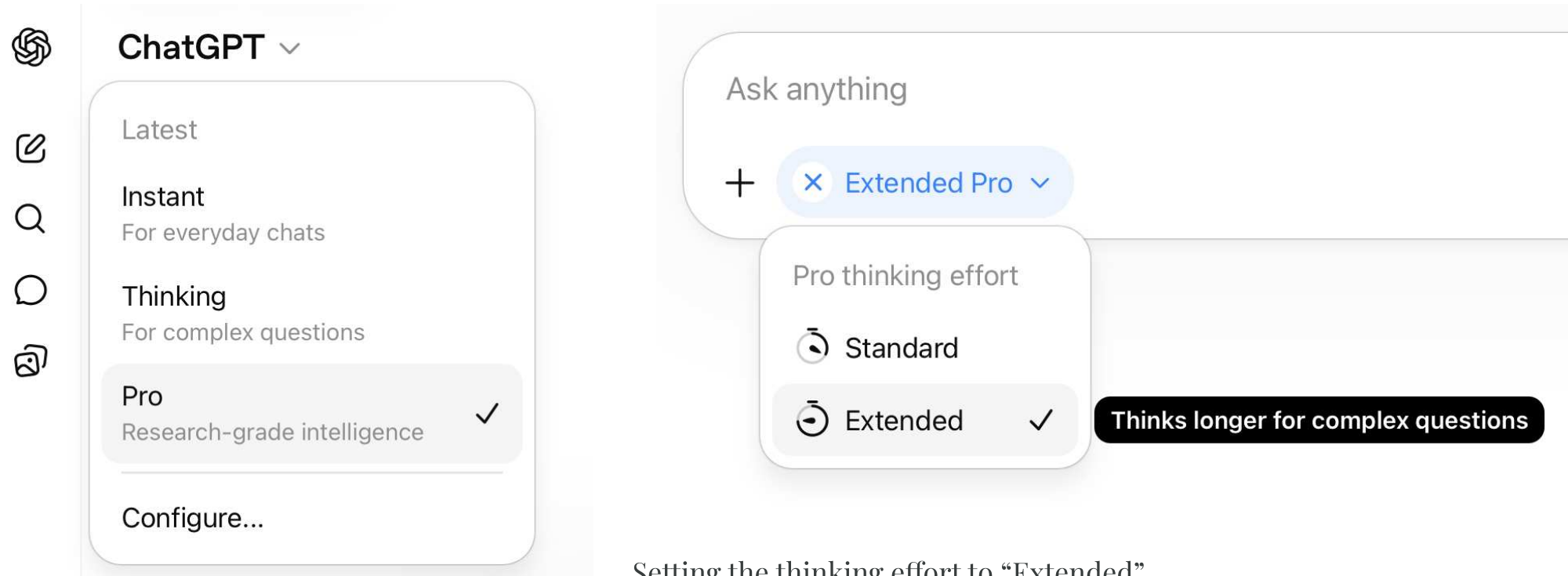


You can get extra usage for free with [GitHub Education for Students](#)

Source: [GitHub Blog](#)

# Reasoning models (2024)

Reasoning is basically automating the old trick of adding “do this step-by-step” to your prompts.



Setting the thinking effort to “Extended”.

Choosing the Pro model.

# Using the larger language models

- Zoom out as far as possible
- Give it as much relevant context as possible
- Better if it's easy to ingest (LaTeX or Python/R code), otherwise it has to convert to text
- A simple instruction in the prompt is enough
- Context sizes for the top models have become quite long
- I've had the best results when it reasons for **20–30 minutes**; then I review each potential issue it finds.

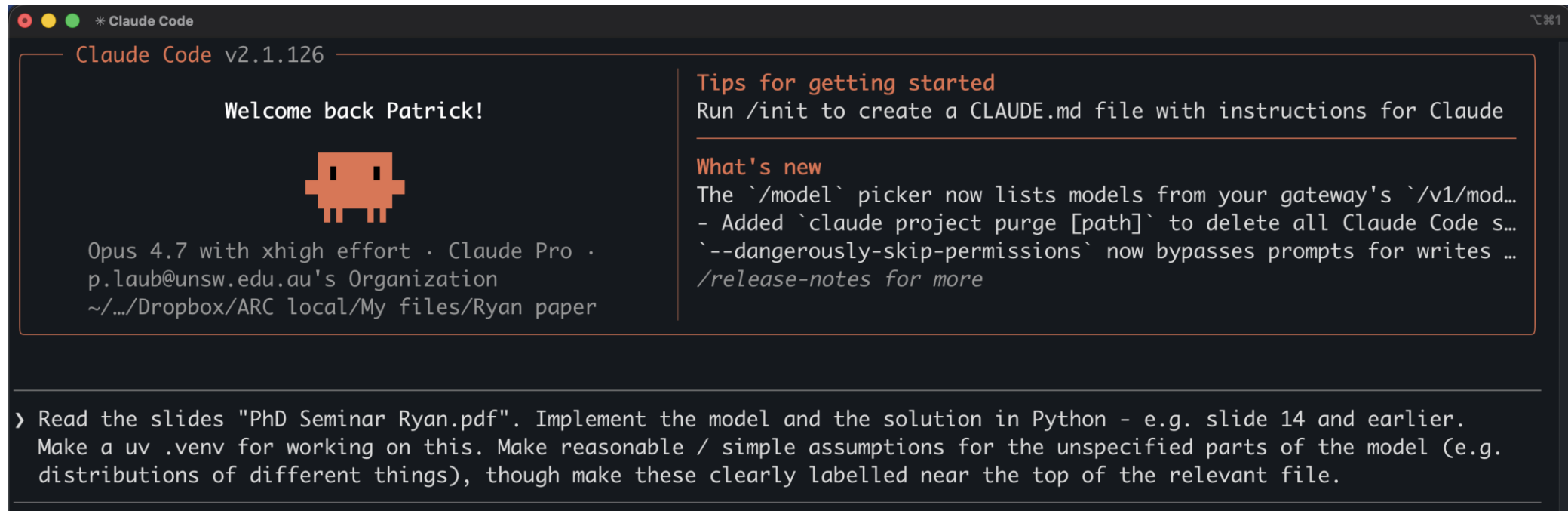
Give me a list of issues (maths, logic, spelling, grammar) in the following manuscript.

```
"""\documentclass[
  journal=aas,
  manuscript=article-type,
  year=2025,
  volume=37,
]{cup-journal}

\usepackage{amsfonts}
\usepackage{amsmath}
\usepackage{amsthm}
\usepackage{graphicx}
\usepackage{subfig}
\usepackage{caption}
\usepackage{xcolor}
```

A typical prompt I use.

# Claude Code (2025)



Let the LLM take control of your terminal/computer.

To get an introduction to using the terminal, check [this recording](#).



# Lecture Outline

- Artificial Intelligence
- Deep Learning Successes
- **Types of Machine Learning Tasks**
- Neural Networks
- California House Price Prediction
- Our First Neural Network
- Force positive predictions
- Preprocessing
- Early Stopping

# Predictive versus generative

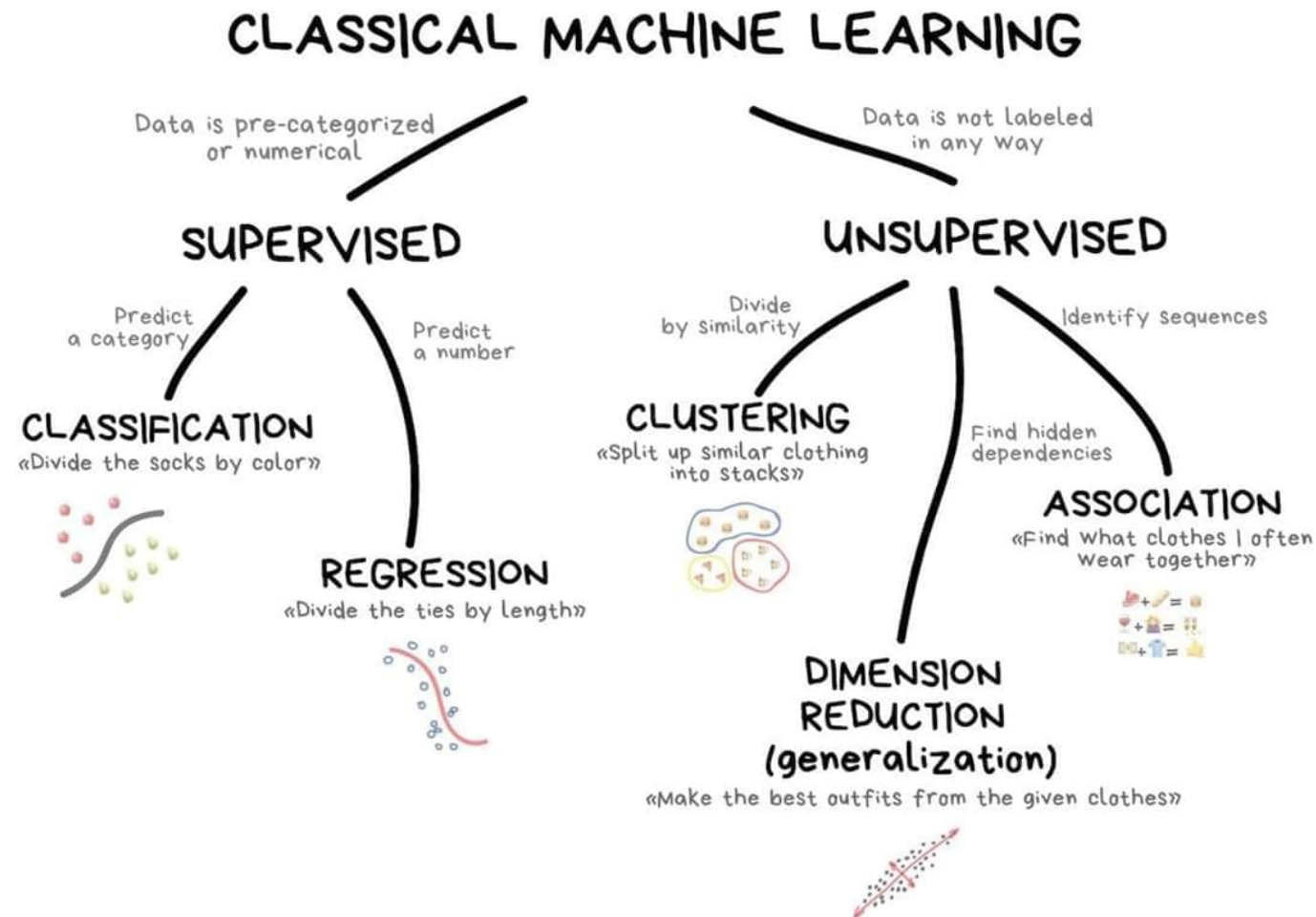
We focus on predictive not generative AI in this course.

Our school has two new courses starting in 2026:

- **ACTL4307** “Generative AI for Actuaries”
- **ACTL4306** “Quantitative Ethical AI for Risk & Actuarial Applications”

This course focuses on **neural networks for supervised learning**, and these techniques are the fundamental building blocks for all the others parts of modern AI.

# A taxonomy of problems



New ones:

- Reinforcement learning
- Semi-supervised learning
- Active learning

Machine learning categories in ACTL3142.

Source: Kaggle, [Getting Started](#).

# Examples of supervised learning

## Regression:

- Given policy  $\hookrightarrow$  predict the rate of claims.
- Given policy  $\hookrightarrow$  predict claim severity.
- Given a reserving triangle  $\hookrightarrow$  predict future claims.

## Classification:

- Given a claim  $\hookrightarrow$  classify as fraudulent or not.
- Given a customer  $\hookrightarrow$  predict customer retention patterns.

# Supervised learning: mathematically

| Background                                                                                                                                                                                                                                                                               | A Recipe for Machine Learning                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>1. Given training data:</p> $\{\mathbf{x}_i, \mathbf{y}_i\}_{i=1}^N$                                                                                                                                                                                                                  | <p>3. Define goal:</p> $\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} \sum_{i=1}^N \ell(f_{\boldsymbol{\theta}}(\mathbf{x}_i), \mathbf{y}_i)$                                                       |
| <p>2. Choose each of these:</p> <ul style="list-style-type: none"> <li>– Decision function</li> </ul> $\hat{\mathbf{y}} = f_{\boldsymbol{\theta}}(\mathbf{x}_i)$ <ul style="list-style-type: none"> <li>– Loss function</li> </ul> $\ell(\hat{\mathbf{y}}, \mathbf{y}_i) \in \mathbb{R}$ | <p>4. Train with SGD:</p> <p>(take small steps opposite the gradient)</p> $\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} - \eta_t \nabla \ell(f_{\boldsymbol{\theta}}(\mathbf{x}_i), \mathbf{y}_i)$ |

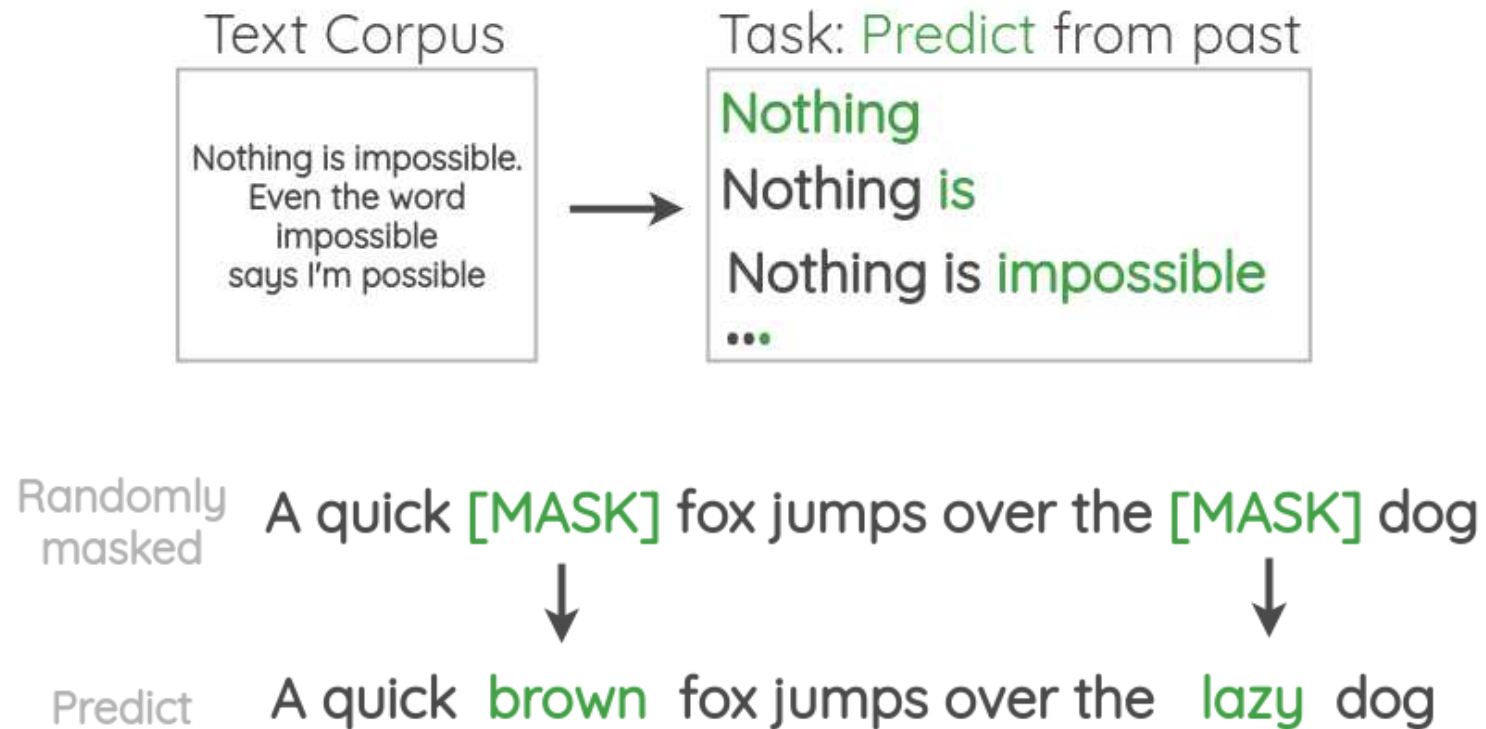
A recipe for supervised learning.

Source: Matthew Gormley (2021), [Introduction to Machine Learning Lecture Slides](#), Slide 67.



# Self-supervised learning

Data which ‘labels itself’. Example: language model.



‘Autoregressive’ (e.g. GPT) versus ‘masked’ model (e.g. BERT).

# Example: image super-resolution



Original image: it is now the target



Downscaled to a lower resolution: it is now the input

Other examples: image inpainting, denoising images.

Test image: Eileen Collins (NASA, public domain).

# Example: Deoldify images



A deoldified version of the famous “Migrant Mother” photograph.

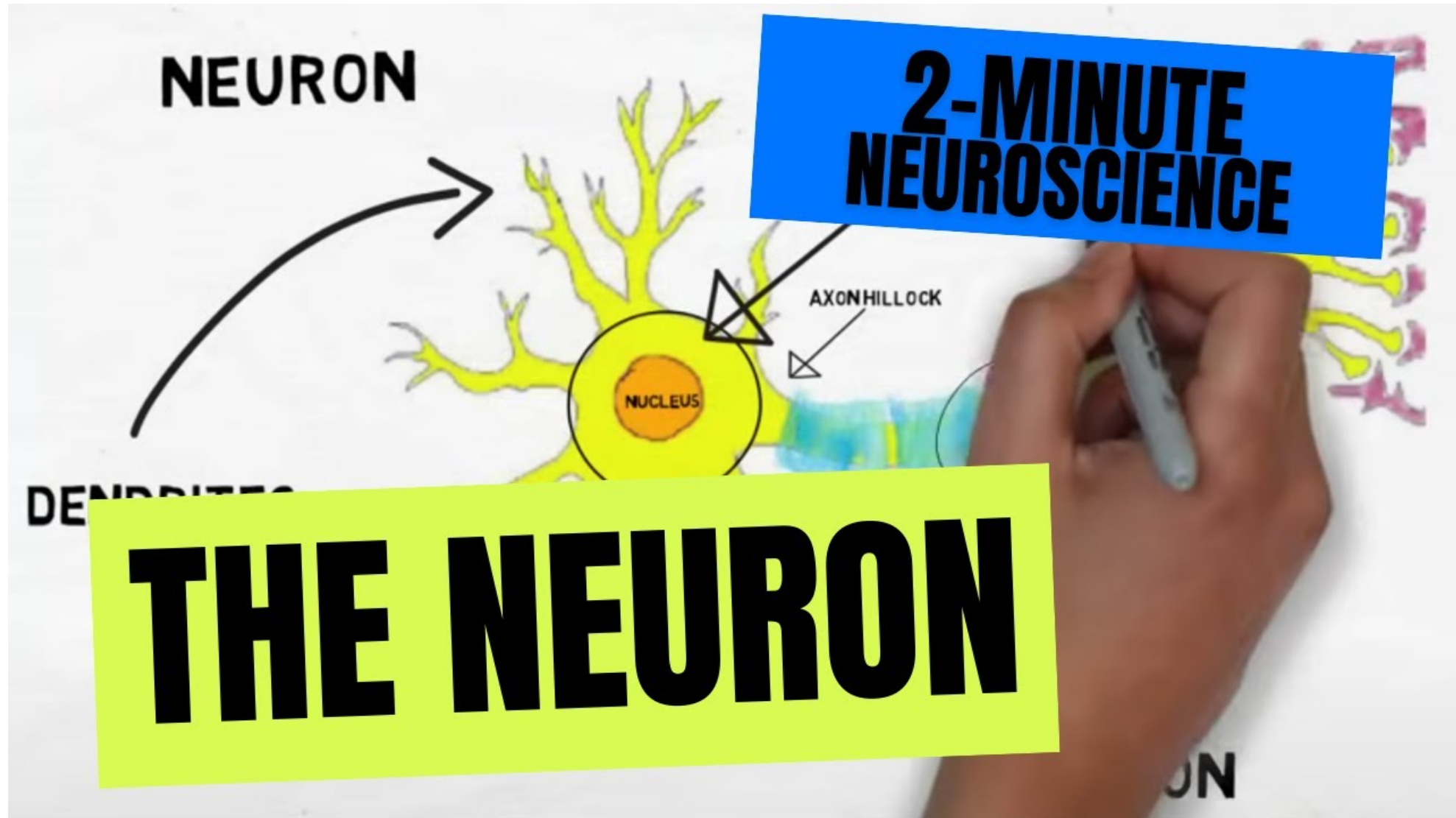
Source: [Deoldify package](#).

# Lecture Outline

- Artificial Intelligence
- Deep Learning Successes
- Types of Machine Learning Tasks
- **Neural Networks**
- California House Price Prediction
- Our First Neural Network
- Force positive predictions
- Preprocessing
- Early Stopping

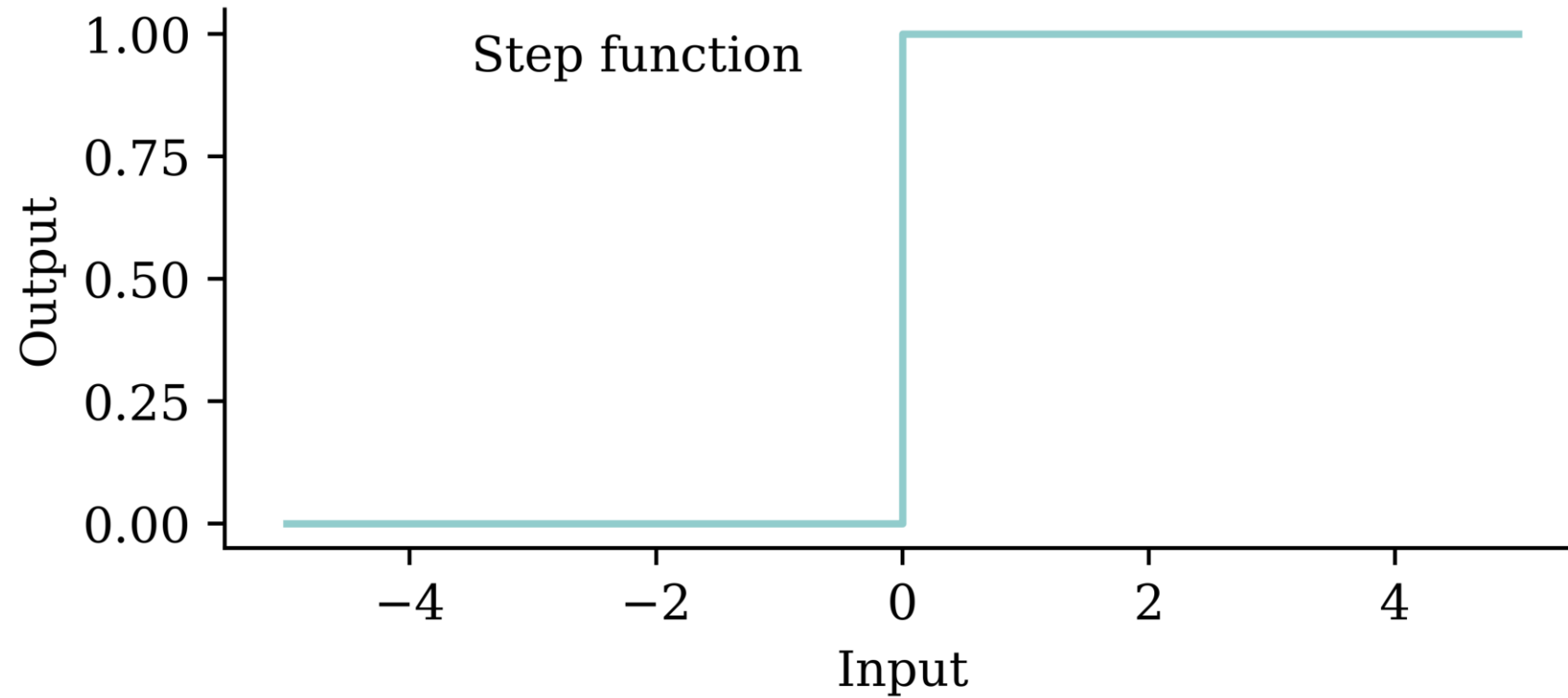


How do real neurons work?

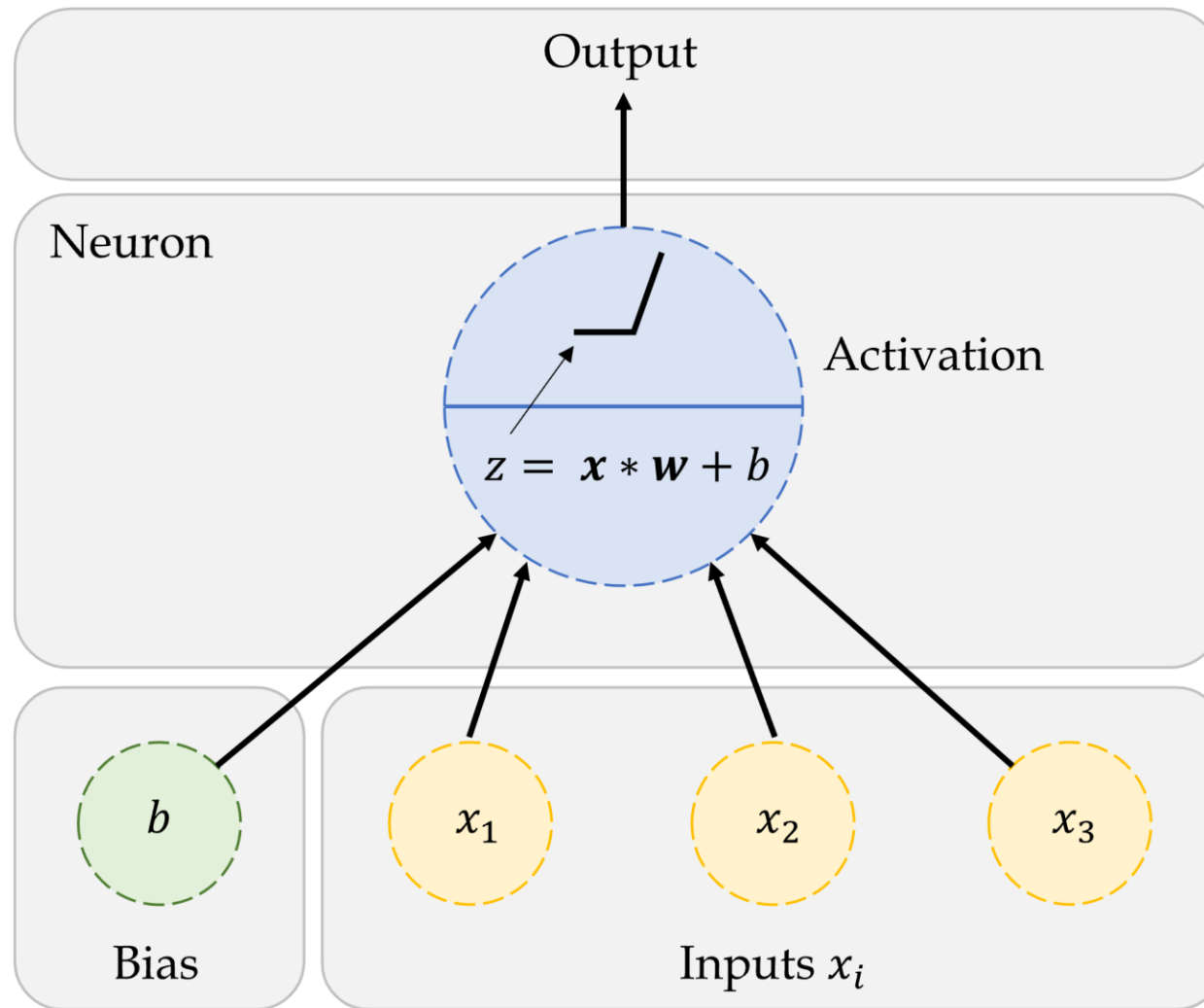




# A neuron 'firing'



# An artificial neuron



A neuron in a neural network with a ReLU activation.

Source: Marcus Lautier (2022).

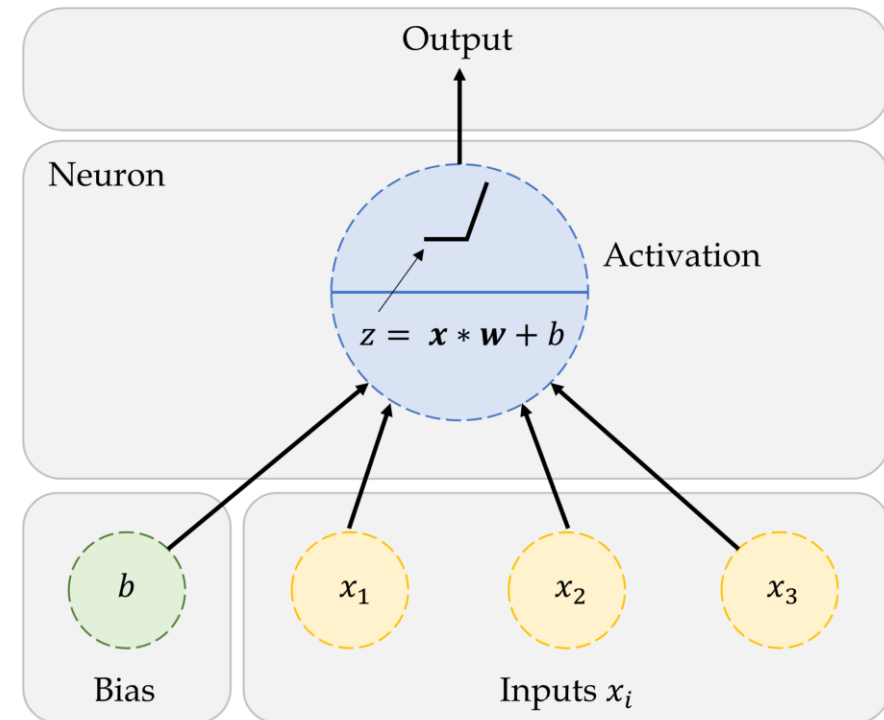
# One neuron

$$z = x_1 \times w_1 + \\ x_2 \times w_2 + \\ x_3 \times w_3.$$

$$a = \begin{cases} z & \text{if } z > 0 \\ 0 & \text{if } z \leq 0 \end{cases}$$

Here,  $x_1, x_2, x_3$  are just some fixed data.

The weights  $w_1, w_2, w_3$  should be ‘learned’.



A neuron in a neural network with a ReLU activation.

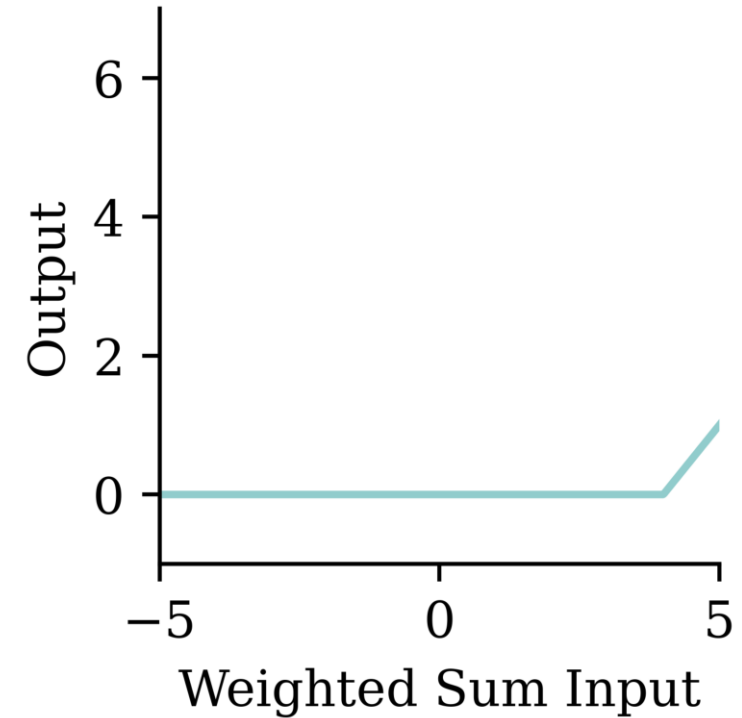
# One neuron with bias

$$z = x_1 \times w_1 + x_2 \times w_2 + x_3 \times w_3 + b.$$

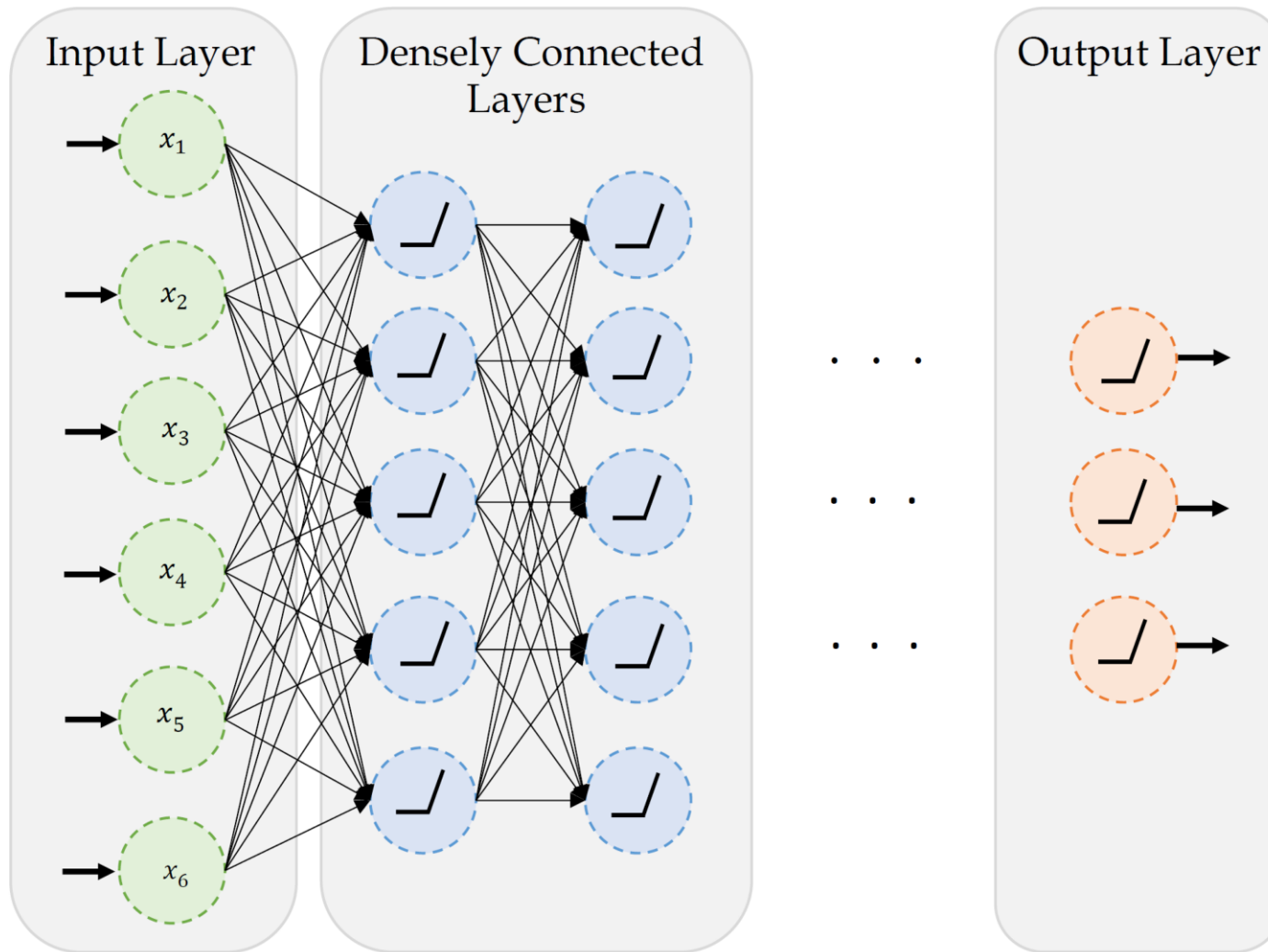
$$a = \begin{cases} z & \text{if } z > 0 \\ 0 & \text{if } z \leq 0 \end{cases}$$

The weights  $w_1$ ,  $w_2$ ,  $w_3$  and bias  $b$  should be ‘learned’.

Bias = -4    0    4



# A basic neural network



A basic fully-connected/dense network.

Source: Marcus Lautier (2022).



# Step-function activation

## Perceptrons

Brains and computers are binary, so make a perceptron with binary data. Seemed reasonable, impossible to train.

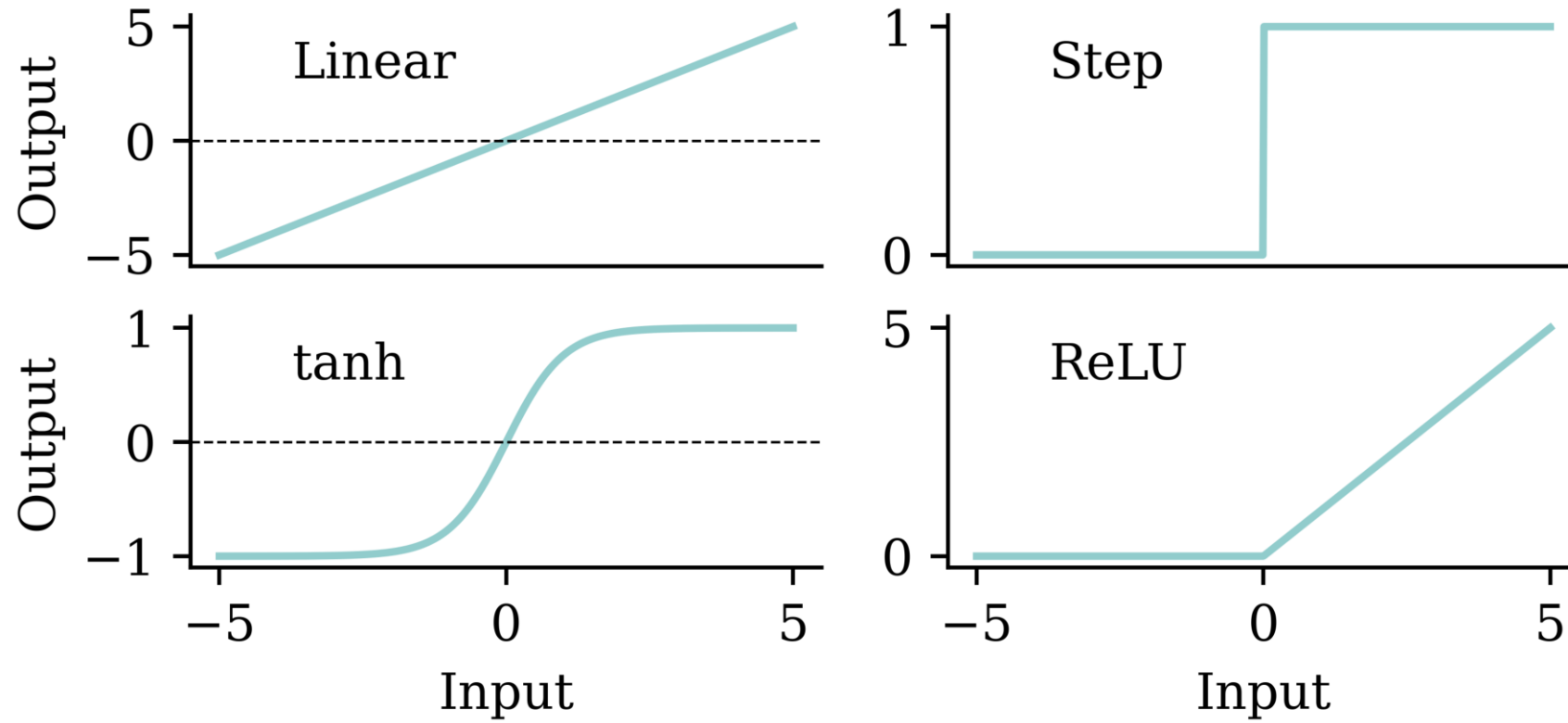
## Modern neural network

Replace binary state with continuous state. Still rather slow to train.

### Note

It's a **neural** network made of **neurons**, not a “neuron network”.

# Try different activation functions

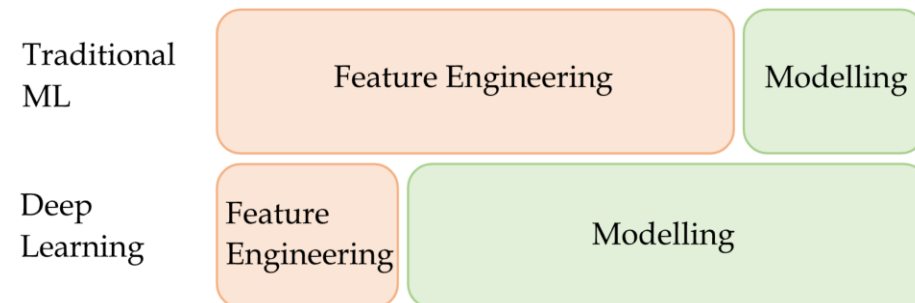
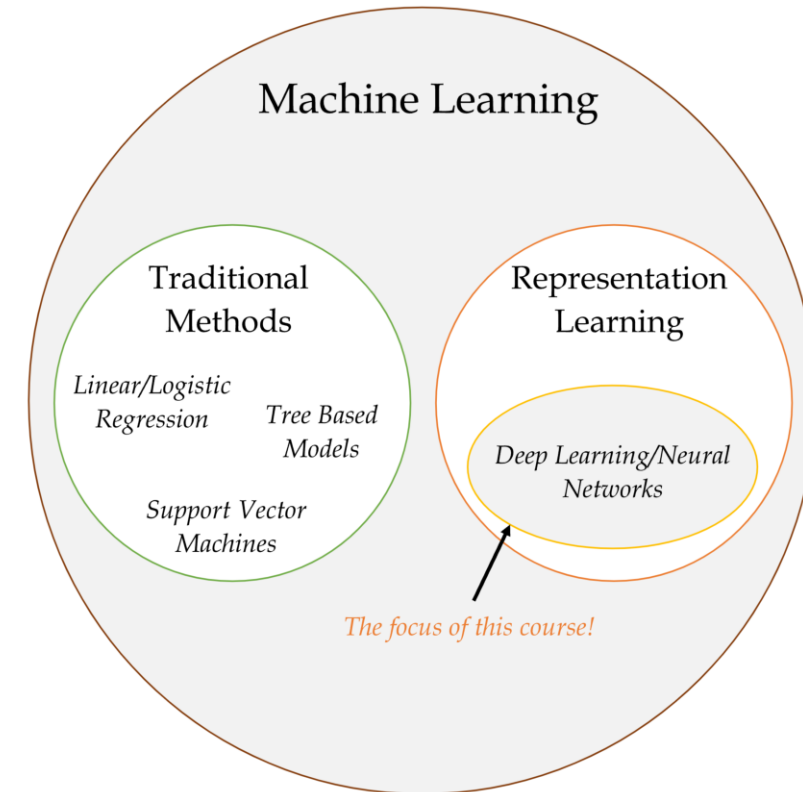
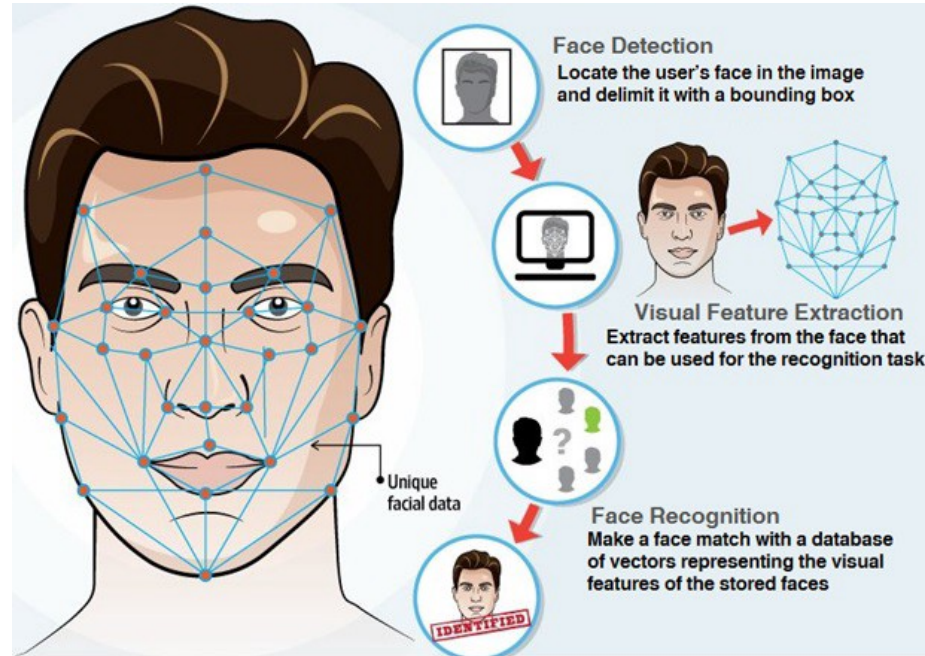


# Flexible

One can show that an MLP is a **universal approximator**, meaning it can model any suitably smooth function, given enough hidden units, to any desired level of accuracy (Hornik 1991). One can either make the model be “wide” or “deep”; the latter has some advantages...

— Murphy (2012, p. 566)

# Feature engineering



Neural networks can learn how to manipulate the inputs (with enough data). That doesn't mean deep learning is always the best option!

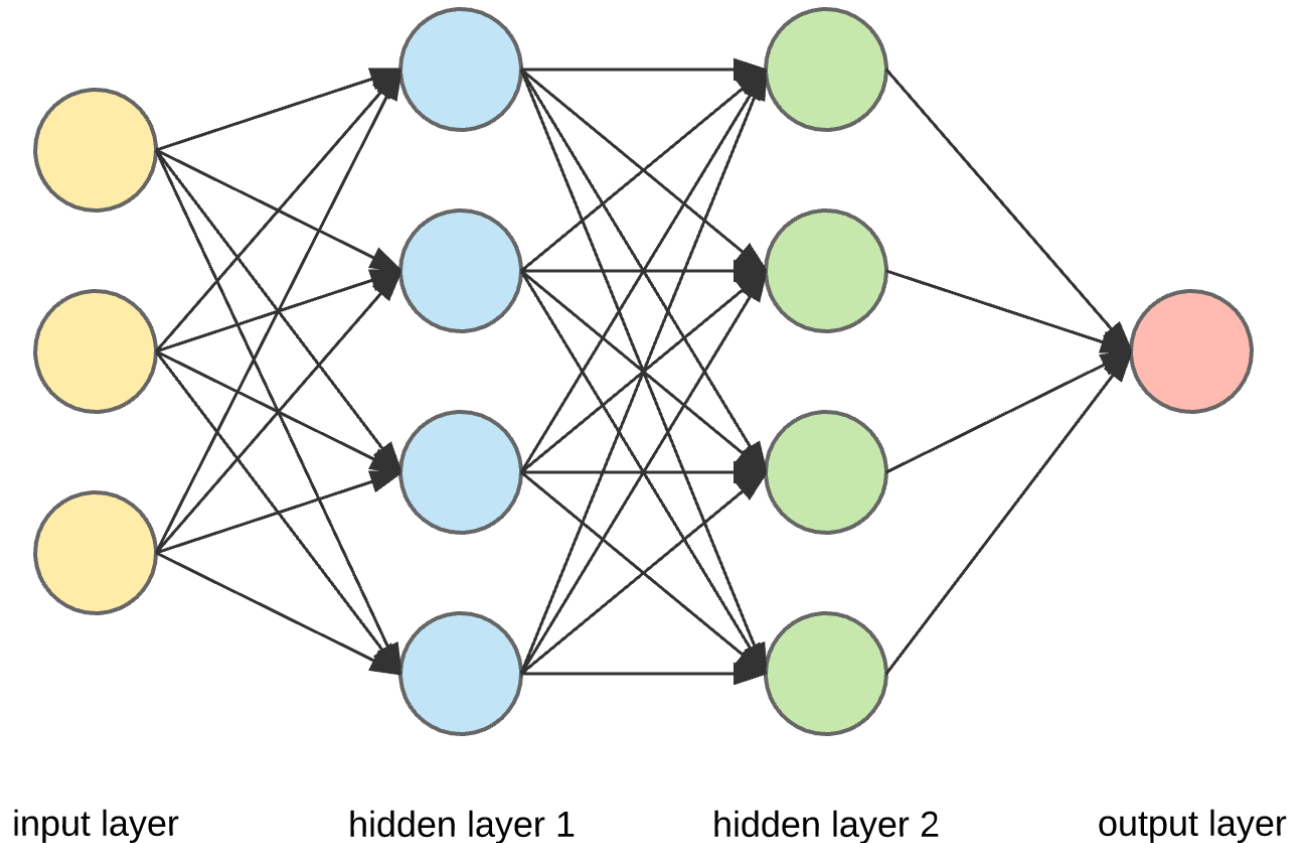
Sources: Marcus Lautier (2022) & Fenjiro (2019), *Face Id: Deep Learning for Face Recognition*, Medium.

# Quiz

In this ANN, how many of the following are there:

- features,
- targets,
- weights,
- biases, and
- parameters?

What is the depth?



An artificial neural network.

# Lecture Outline

- Artificial Intelligence
- Deep Learning Successes
- Types of Machine Learning Tasks
- Neural Networks
- **California House Price Prediction**
- Our First Neural Network
- Force positive predictions
- Preprocessing
- Early Stopping

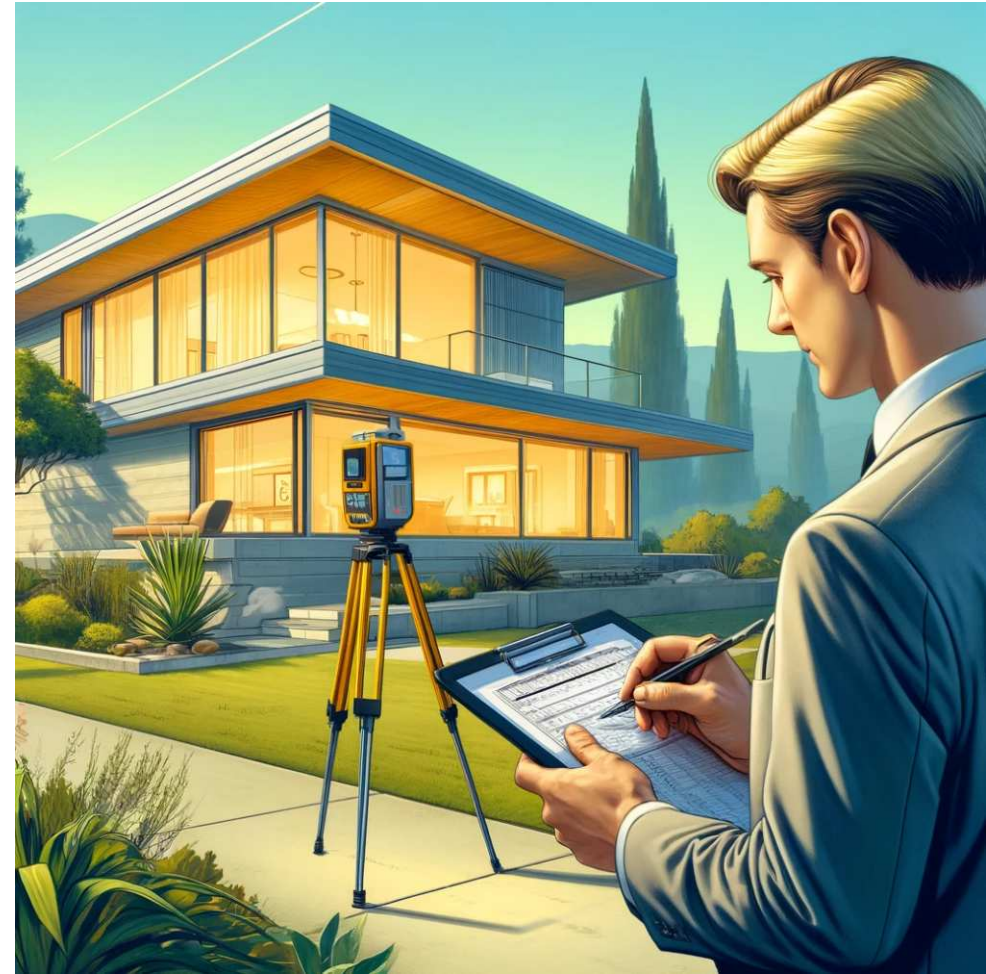


# Imports needed for this demo

```
1 import random
2
3 import numpy as np
4 import pandas as pd
5
6 from sklearn.datasets import fetch_california_housing
7 from sklearn.model_selection import train_test_split
8 from sklearn.linear_model import LinearRegression
9 from sklearn.metrics import mean_squared_error
10 from sklearn.preprocessing import StandardScaler, MinMaxScaler
11
12 from keras.models import Sequential
13 from keras.layers import Dense, Input
14 from keras.callbacks import EarlyStopping
15 from keras.callbacks import ModelCheckpoint
```

# Data science always starts with the data!

The target variable is the median house value for California districts, expressed in \$100,000's. This dataset was derived from the 1990 U.S. census, using one row per census block group. A block group is the smallest geographical unit for which the U.S. Census Bureau publishes sample data (a block group typically has a population of 600 to 3,000 people).



Dall-E's rendition of this dataset.

Source: [Scikit-learn documentation](#).

# Columns

- `MedInc` median income in block group
- `HouseAge` median house age in block group
- `AveRooms` average number of rooms per household
- `AveBedrms` average # of bedrooms per household
- `Population` block group population
- `AveOccup` average number of household members
- `Latitude` block group latitude
- `Longitude` block group longitude
- `MedHouseVal` median house value (**target**)

# Import the data

```
1 features, target = fetch_california_housing(as_frame=True, return_X_y=True)
2 features
```

|              | MedInc | HouseAge | AveRooms | AveBedrms | Population | AveOccup | Latitude | Longitude |
|--------------|--------|----------|----------|-----------|------------|----------|----------|-----------|
| <b>0</b>     | 8.3252 | 41.0     | 6.984127 | 1.023810  | 322.0      | 2.555556 | 37.88    | -122.23   |
| <b>1</b>     | 8.3014 | 21.0     | 6.238137 | 0.971880  | 2401.0     | 2.109842 | 37.86    | -122.22   |
| <b>2</b>     | 7.2574 | 52.0     | 8.288136 | 1.073446  | 496.0      | 2.802260 | 37.85    | -122.24   |
| ...          | ...    | ...      | ...      | ...       | ...        | ...      | ...      | ...       |
| <b>20637</b> | 1.7000 | 17.0     | 5.205543 | 1.120092  | 1007.0     | 2.325635 | 39.43    | -121.22   |
| <b>20638</b> | 1.8672 | 18.0     | 5.329513 | 1.171920  | 741.0      | 2.123209 | 39.43    | -121.32   |
| <b>20639</b> | 2.3886 | 16.0     | 5.254717 | 1.162264  | 1387.0     | 2.616981 | 39.37    | -121.24   |

20640 rows × 8 columns

# Train/validation/test split

```
1 X_main, X_test, y_main, y_test = train_test_split(  
2     features, target, test_size=0.2, random_state=1)  
3 X_train, X_val, y_train, y_val = train_test_split(  
4     X_main, y_main, test_size=0.25, random_state=1)  
5  
6 num_features = features.shape[1]  
7 print(X_train.shape, X_val.shape, X_test.shape)
```

(12384, 8) (4128, 8) (4128, 8)

# Linear regression baseline

Refit the linear regression from earlier; we'll compare neural networks against this baseline.

```
1 lr = LinearRegression()  
2 lr.fit(X_train, y_train)  
3  
4 mse_lr_train = mean_squared_error(y_train, lr.predict(X_train))  
5 mse_lr_val = mean_squared_error(y_val, lr.predict(X_val))  
6  
7 mse_train = {"Linear Regression": mse_lr_train}  
8 mse_val = {"Linear Regression": mse_lr_val}
```

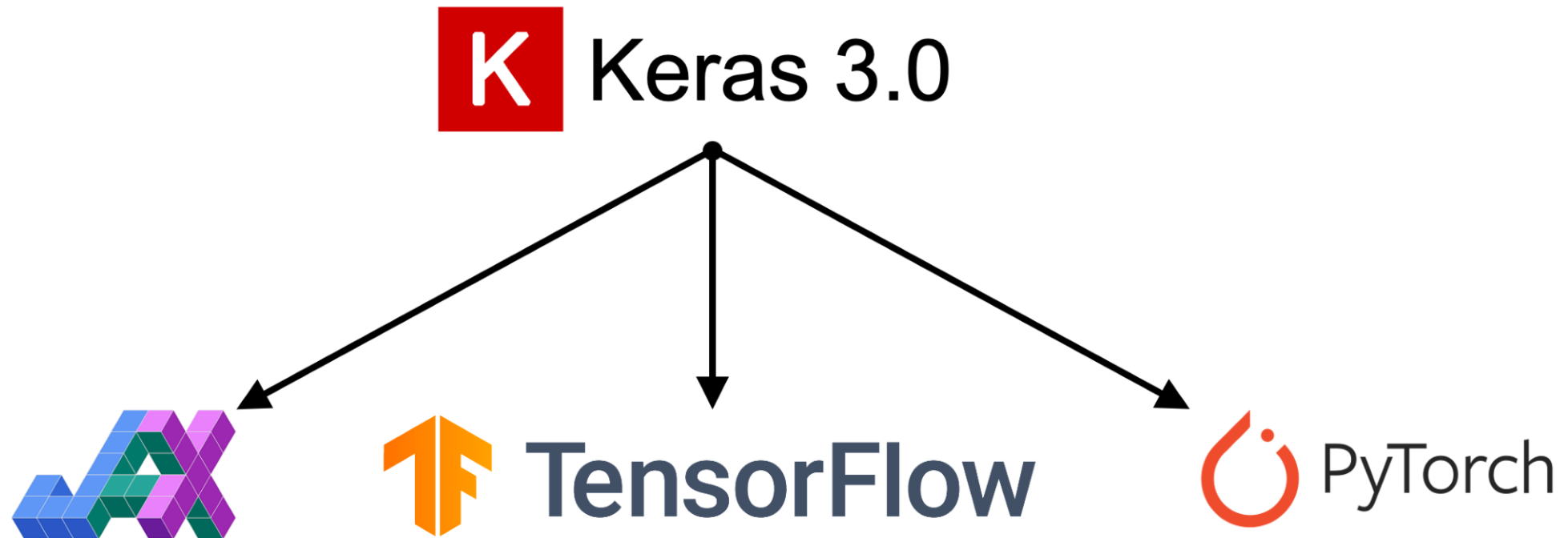


# Lecture Outline

- Artificial Intelligence
- Deep Learning Successes
- Types of Machine Learning Tasks
- Neural Networks
- California House Price Prediction
- **Our First Neural Network**
- Force positive predictions
- Preprocessing
- Early Stopping

# What are Keras and PyTorch?

Keras is a common way of specifying, training, and using neural networks. It gives a simple interface to *various backend* libraries, including PyTorch.



The Keras application programming interface (API)

Source: Melissa Renard (2025)

# Create a Keras ANN model

Decide on the architecture: a simple fully-connected network with one hidden layer with 30 neurons.

Create the model:

```
1 model = Sequential(  
2     [Input((num_features,)),  
3     Dense(30, activation="leaky_relu"),  
4     Dense(1, activation="leaky_relu")]  
5 )
```

# Inspect the model

```
1 model.summary()
```

Model: "sequential"

| Layer (type)    | Output Shape | Param # |
|-----------------|--------------|---------|
| dense (Dense)   | (None, 30)   | 270     |
| dense_1 (Dense) | (None, 1)    | 31      |

Total params: 301 (1.18 KB)

Trainable params: 301 (1.18 KB)

Non-trainable params: 0 (0.00 B)

# The model is initialised randomly

```
1 model = Sequential([Dense(30, activation="leaky_relu"), Dense(1, activation="leaky_relu")  
2 model.predict(X_val.head(3), verbose=0)
```

```
array([[ -139.05],  
       [  -84.57],  
       [   -5.82]], dtype=float32)
```

```
1 model = Sequential([Dense(30, activation="leaky_relu"), Dense(1, activation="leaky_relu")  
2 model.predict(X_val.head(3), verbose=0)
```

```
array([[ -108.21],  
       [  -64.74],  
       [   -7.1 ]], dtype=float32)
```

# Controlling the randomness

```
1 random.seed(2026)
2
3 model = Sequential([Dense(30, activation="leaky_relu"), Dense(1, activation="leaky_relu")
4
5 display(model.predict(X_val.head(3), verbose=0))
6
7 random.seed(2026)
8 model = Sequential([Dense(30, activation="leaky_relu"), Dense(1, activation="leaky_relu")
9
10 display(model.predict(X_val.head(3), verbose=0))
```

```
array([[467.02],
       [289.13],
       [ 20.93]], dtype=float32)
```

```
array([[467.02],
       [289.13],
       [ 20.93]], dtype=float32)
```



# Fit the model

```
1 random.seed(2026)
2
3 model = Sequential([
4     Dense(30, activation="leaky_relu"),
5     Dense(1, activation="leaky_relu")
6 ])
7
8 model.compile("adam", "mse")
9 %time hist = model.fit(X_train, y_train, epochs=5, verbose=0)
10 hist.history["loss"]
```

CPU times: user 3.57 s, sys: 111 ms, total: 3.68 s

Wall time: 3.62 s

```
[1095.7493896484375,
 6.3134589195251465,
 4.662435531616211,
 3.2442092895507812,
 1.773606300354004]
```

# Make predictions

```
1 y_pred = model.predict(X_train[:3], verbose=0)
2 y_pred
```

```
array([[ 1.74],
       [-0.83],
       [ 1.77]], dtype=float32)
```

## Note

The `.predict` gives us a 'matrix' not a 'vector'. Calling `.flatten()` will convert it to a 'vector'.

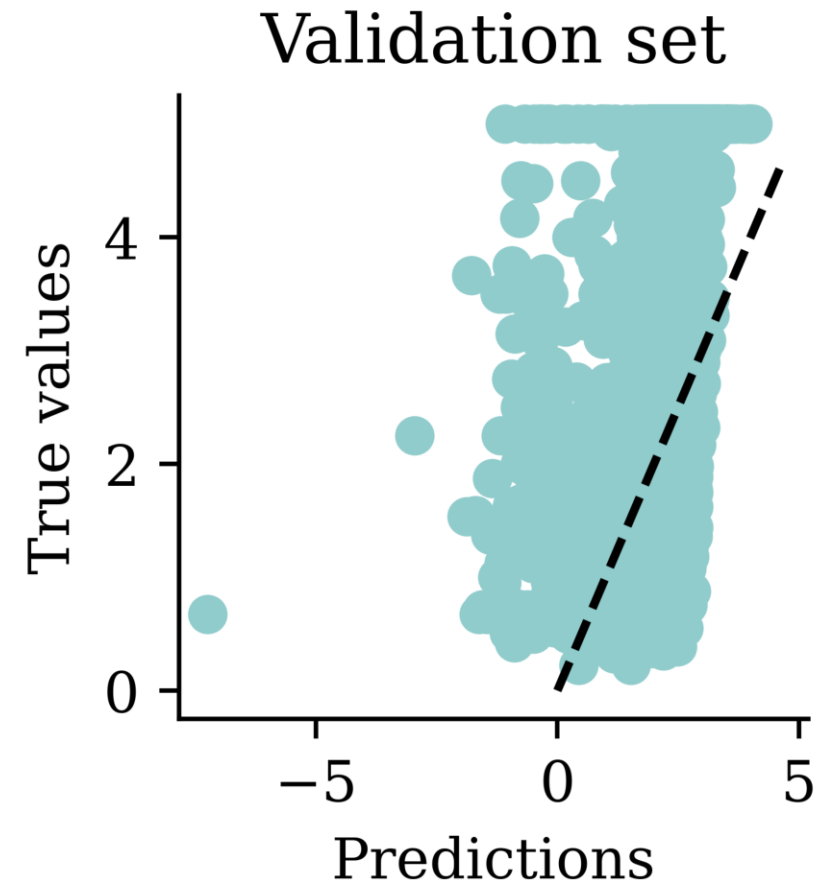
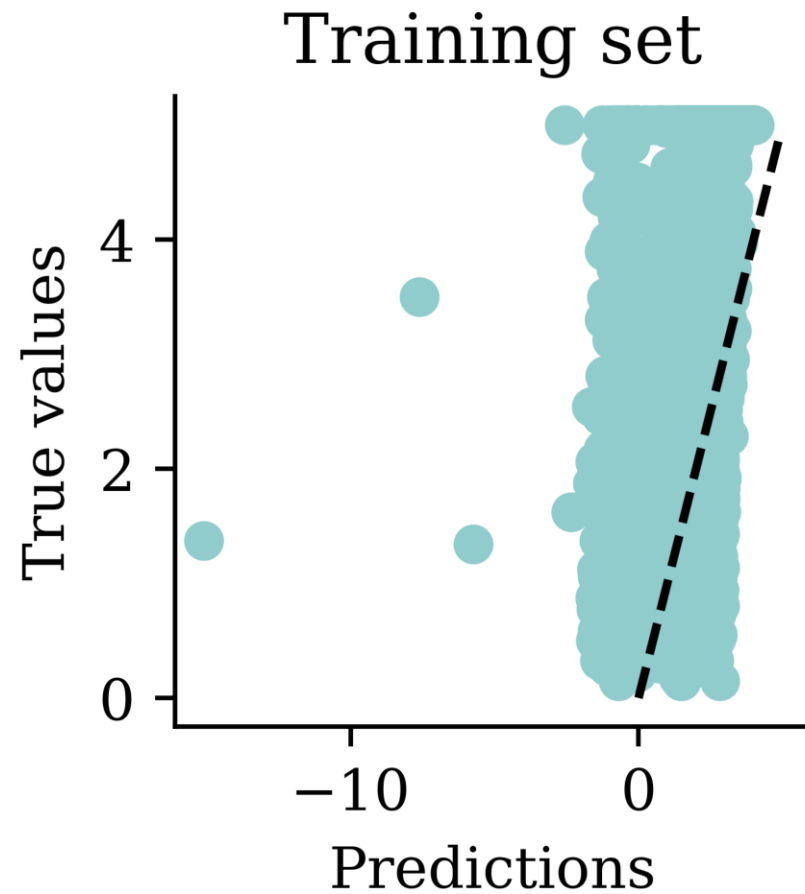
```
1 print(f"Original shape: {y_pred.shape}")
2 y_pred = y_pred.flatten()
3 print(f"Flattened shape: {y_pred.shape}")
4 y_pred
```

Original shape: (3, 1)

Flattened shape: (3,)

```
array([ 1.74, -0.83,  1.77], dtype=float32)
```

# Plot the predictions



# Assess the model

```
1 y_pred = model.predict(X_val, verbose=0)
2 mean_squared_error(y_val, y_pred)
```

1.322503585825187

```
1 mse_train["Basic ANN"] = mean_squared_error(
2     y_train, model.predict(X_train, verbose=0)
3 )
4 mse_val["Basic ANN"] = mean_squared_error(y_val, model.predict(X_val, verbose=0))
```

Some predictions are negative:

```
1 y_pred = model.predict(X_val, verbose=0)
2 y_pred.min(), y_pred.max()
```

(np.float32(-7.2446113), np.float32(4.038248))

```
1 y_val.min(), y_val.max()
```

(np.float64(0.225), np.float64(5.00001))

# Lecture Outline

- Artificial Intelligence
- Deep Learning Successes
- Types of Machine Learning Tasks
- Neural Networks
- California House Price Prediction
- Our First Neural Network
- **Force positive predictions**
- Preprocessing
- Early Stopping

# Try running for longer

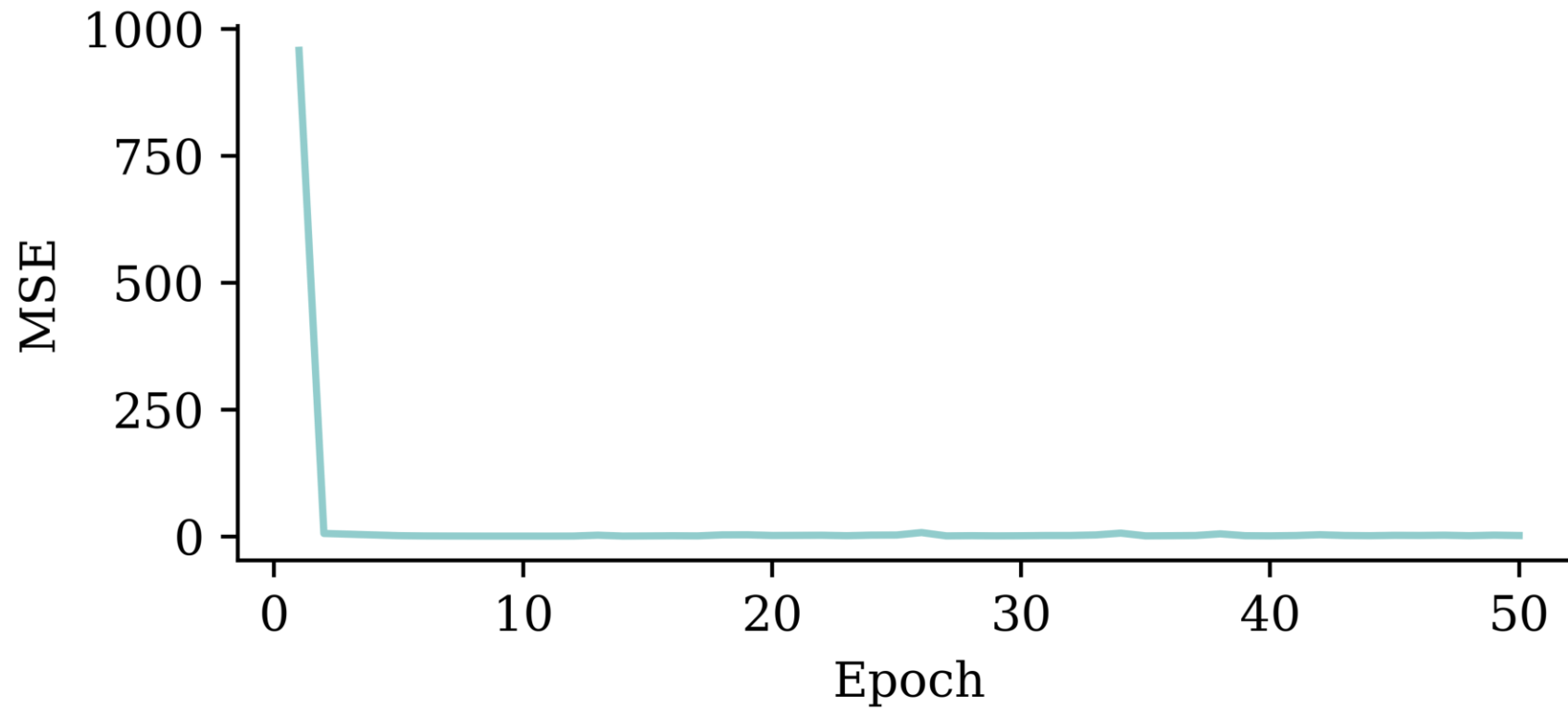
```
1 random.seed(2026)
2
3 model = Sequential([
4     Dense(30, activation="leaky_relu"),
5     Dense(1, activation="leaky_relu")
6 ])
7
8 model.compile("adam", "mse")
9
10 %time hist = model.fit(X_train, y_train, epochs=50, verbose=0)
```

CPU times: user 33.7 s, sys: 787 ms, total: 34.5 s

Wall time: 33.9 s

# Loss curve

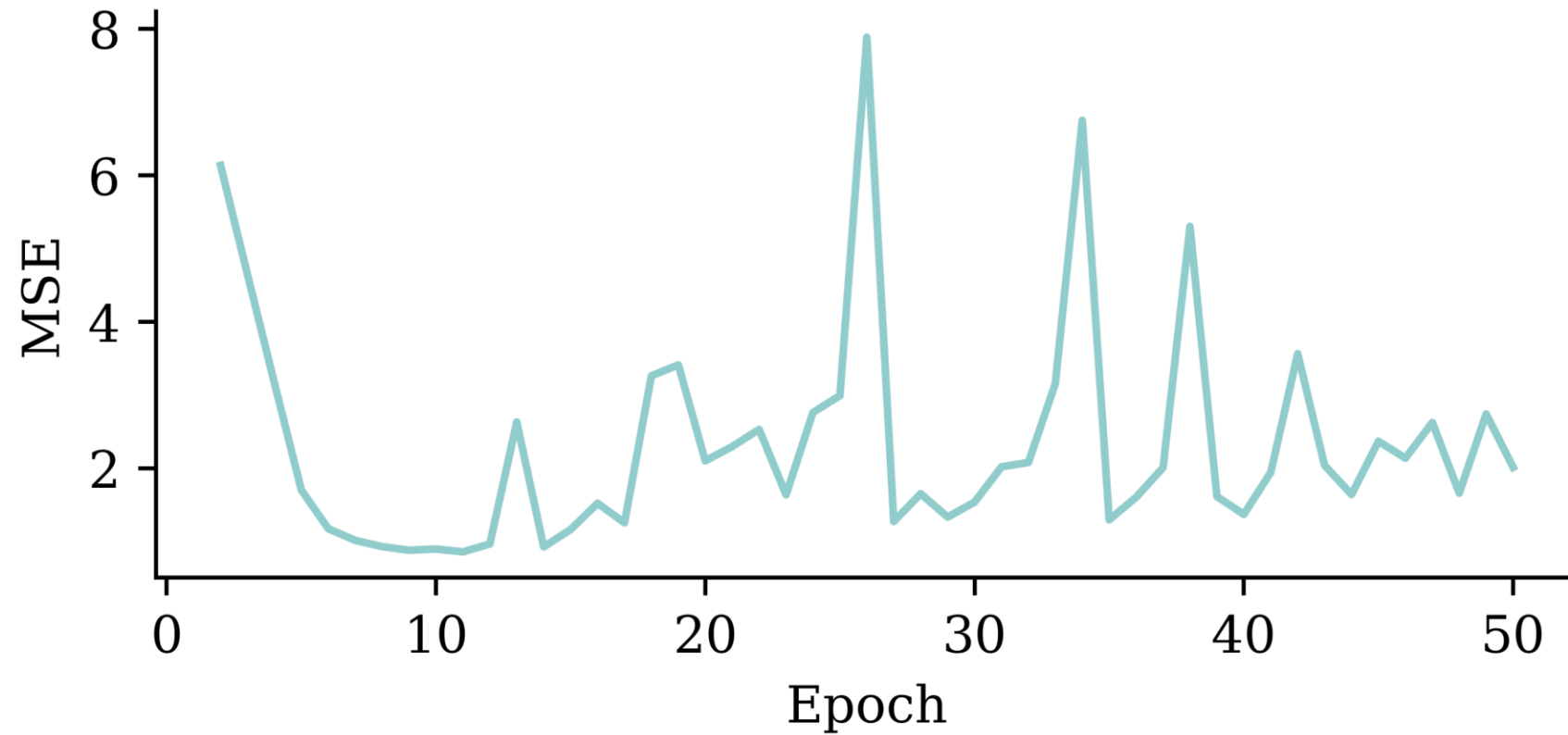
```
1 plt.plot(range(1, 51), hist.history["loss"])
2 plt.xlabel("Epoch")
3 plt.ylabel("MSE");
```





# Loss curve

```
1 plt.plot(range(2, 51), hist.history["loss"][1:])  
2 plt.xlabel("Epoch")  
3 plt.ylabel("MSE");
```



# Predictions

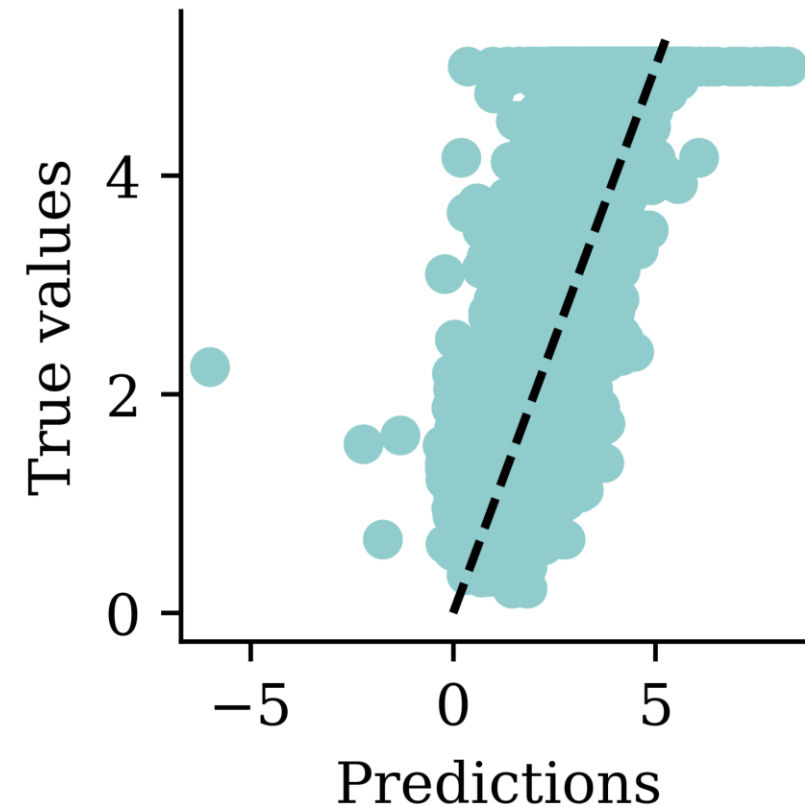
```
1 y_pred = model.predict(X_val, verbose=0)
2 print(f"Min prediction: {y_pred.min():.2f}")
3 print(f"Max prediction: {y_pred.max():.2f}")
```

Min prediction: -6.00

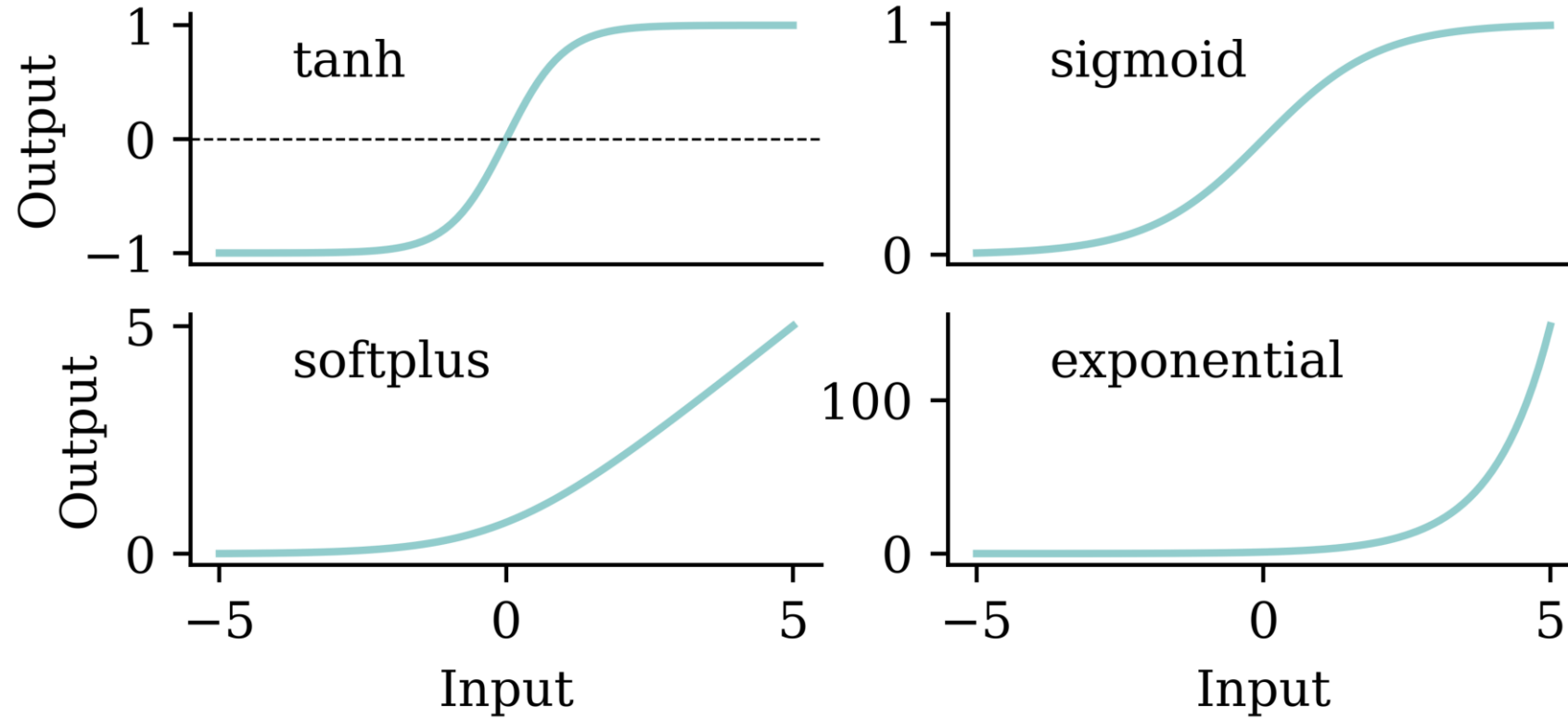
Max prediction: 8.25

```
1 plt.scatter(y_pred, y_val)
2 plt.xlabel("Predictions")
3 plt.ylabel("True values")
4 add_diagonal_line()
```

```
1 mse_train["Long run ANN"] = mean_squared_
2     y_train, model.predict(X_train, verbo
3 )
4 mse_val["Long run ANN"] = mean_squared_er
```



# Try different activation functions



# Enforce positive outputs (softplus)

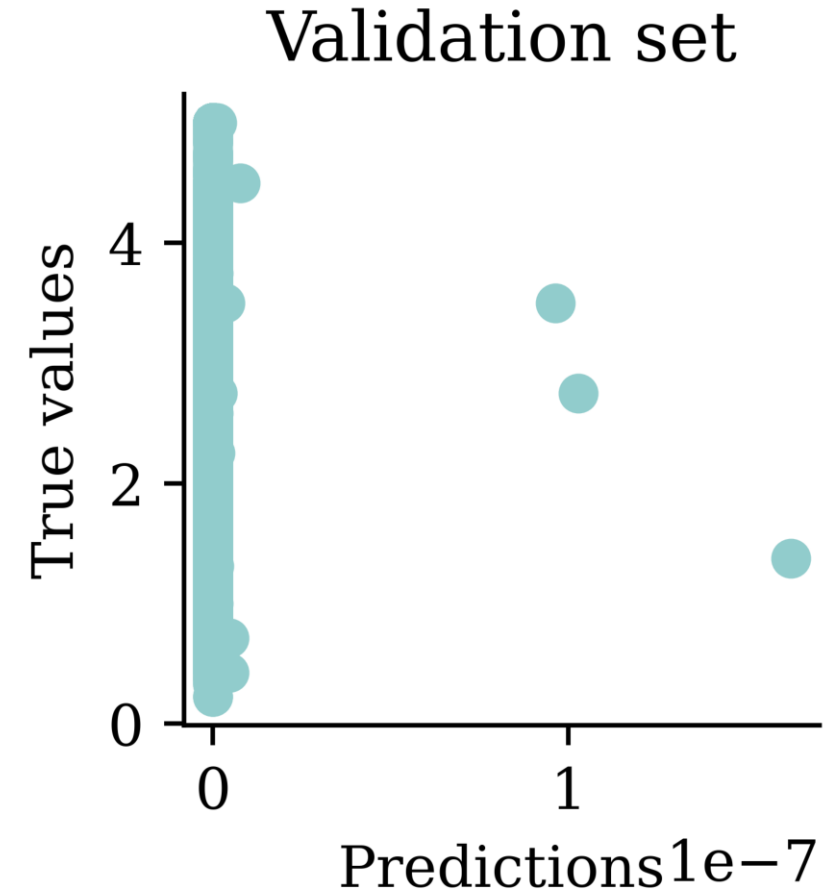
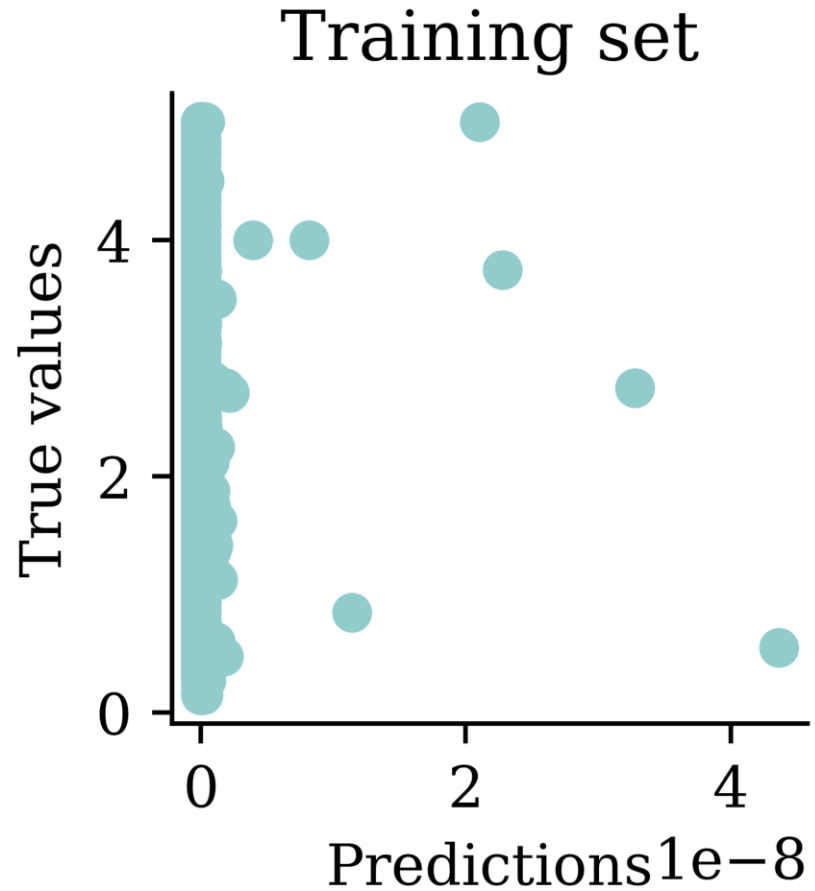
```
1 random.seed(2026)
2
3 model = Sequential([
4     Dense(30, activation="leaky_relu"),
5     Dense(1, activation="softplus")
6 ])
7
8 model.compile("adam", "mse")
9
10 %time hist = model.fit(X_train, y_train, epochs=50, verbose=0)
11
12 losses = np.round(hist.history["loss"], 2)
13 print(losses[:5], " ... ", losses[-5:])
```

CPU times: user 33.9 s, sys: 822 ms, total: 34.7 s

Wall time: 34.1 s

[973.53 5.64 5.64 5.64 5.64] ... [5.64 5.64 5.64 5.64 5.64]

# Plot the predictions



# Enforce positive outputs ( $e^x$ )

```
1 random.seed(2026)
2
3 model = Sequential([
4     Dense(30, activation="leaky_relu"),
5     Dense(1, activation="exponential")
6 ])
7
8 model.compile("adam", "mse")
9
10 %time hist = model.fit(X_train, y_train, epochs=5, verbose=0)
11
12 losses = hist.history["loss"]
13 print(losses)
```

CPU times: user 3.43 s, sys: 77.2 ms, total: 3.51 s

Wall time: 3.45 s

[nan, nan, nan, nan, nan]

# Same as transforming the target

## 4.1. Model

We fitted the following model:

$$\begin{aligned} \ln(\text{MEDIAN VALUE}) = & \text{INTERCEPT} + \beta_2 \text{ MEDIAN INCOME} + \beta_3 \text{ MEDIAN INCOME}^2 + \beta_4 \text{ MEDIAN INCOME}^3 \\ & + \beta_5 \ln(\text{MEDIAN(AGE)}) + \beta_6 \ln(\text{TOTAL ROOMS/POPULATION}) \\ & + \beta_7 \ln(\text{BEDROOMS/POPULATION}) + \beta_8 \ln(\text{POPULATION/HOUSEHOLDS}) \\ & + \beta_9 \ln(\text{HOUSEHOLDS}) \end{aligned} \quad (8)$$

The polynomial regression used by researchers who first studied this dataset.

### Note

Fitting  $\ln(\text{Median Value})$  is mathematically identical to the [exponential](#) activation function in the final layer (but metrics are in different units).



# Good to know others results

Table 1

OLS and SAR estimates for median housing prices across 20 640 California census block groups

|                                  | $B_{ols}$ | $t_{ols}$ | $B_{sar}$ | $t_{sar}$ |
|----------------------------------|-----------|-----------|-----------|-----------|
| INTERCEPT                        | 11.4939   | 275.7518  | 11.6637   | 402.5925  |
| MEDIAN INCOME                    | 0.4790    | 45.7768   | 0.0349    | 4.7104    |
| MEDIAN INCOME <sup>2</sup>       | -0.0166   | -9.4841   | 0.0100    | 8.4280    |
| MEDIAN INCOME <sup>3</sup>       | -0.0002   | -1.9157   | -0.0007   | -12.2444  |
| ln(MEDIAN AGE)                   | 0.1570    | 33.6123   | -0.0421   | -11.0942  |
| ln(TOTAL ROOMS/POPULATION)       | -0.8582   | -56.1280  | 0.3098    | 24.5768   |
| ln(BEDROOMS/POPULATION)          | 0.8043    | 38.0685   | -0.1926   | -11.8049  |
| ln(POPULATION/HOUSEHOLDS)        | -0.4077   | -20.8762  | -0.0342   | -2.3582   |
| ln(HOUSEHOLDS)                   | 0.0477    | 13.0792   | 0.0034    | 1.5569    |
| $\alpha$                         |           |           | 0.8536    |           |
| $R^2$                            | 0.6078    |           | 0.8594    |           |
| Median $ e $                     | 0.2101    |           | 0.1084    |           |
| Execution time                   |           |           | 1130 s    |           |
| Number of likelihood evaluations |           |           | 10        |           |

That basic model gets  $R^2$  of 0.61, but their fancy model gets 0.86.

Source: Pace and Barry (1997), [Sparse Spatial Autoregressions](#), Statistics & Probability Letters.

# GPT can double-check these results

We fitted the following model:

$$\ln(\text{MEDIAN VALUE}) = \text{INTERCEPT} + \beta_2 \text{ MEDIAN INCOME} + \beta_3 \text{ MEDIAN INCOME}^2 + \beta_4 \text{ MEDIAN INCOME}^3 \\ + \beta_5 \ln(\text{MEDIAN(AGE)}) + \beta_6 \ln(\text{TOTAL ROOMS/POPULATION}) \\ + \beta_7 \ln(\text{BEDROOMS/POPULATION}) + \beta_8 \ln(\text{POPULATION/HOUSEHOLDS}) \\ + \beta_9 \ln(\text{HOUSEHOLDS}) \quad (8)$$

Fit the linear model from this screenshot to the data using Python, and calculate the  $R^2$

The fitted linear model as per the screenshot yields an  $R^2$  value of approximately 0.608. This indicates that approximately 60.8% of the variance in the log of median house value can be explained by the model. [↗](#)

Asking GPT to check it.

I'd previously given it the CSV of the data.

```
python
import numpy as np
import statsmodels.api as sm

# Prepare the data
cal_housing['MedInc2'] = cal_housing['MedInc'] ** 2
cal_housing['MedInc3'] = cal_housing['MedInc'] ** 3
cal_housing['log_HouseAge'] = np.log(cal_housing['HouseAge'])
cal_housing['log_AveRooms_Population'] = np.log(cal_housing['AveRooms'] /
cal_housing['Population'])
cal_housing['log_AveBedrms_Population'] = np.log(cal_housing['AveBedrms'] /
cal_housing['Population'])
cal_housing['log_Population_Households'] = np.log(cal_housing['Population'] /
cal_housing['AveOccup'])
cal_housing['log_AveOccup'] = np.log(cal_housing['AveOccup'])

# Define the target variable and the predictors
y = np.log(cal_housing['MedHouseVal'])
X = cal_housing[['MedInc', 'MedInc2', 'MedInc3', 'log_HouseAge',
'log_AveRooms_Population',
'log_AveBedrms_Population', 'log_Population_Households',
'log_AveOccup']]

# Add a constant to the model (the intercept term)
X = sm.add_constant(X)

# Fit the model
model = sm.OLS(y, X).fit()

# Get the R-squared value
r_squared = model.rsquared
r_squared

Result
0.6077752947111696
```

The code it wrote & ran.

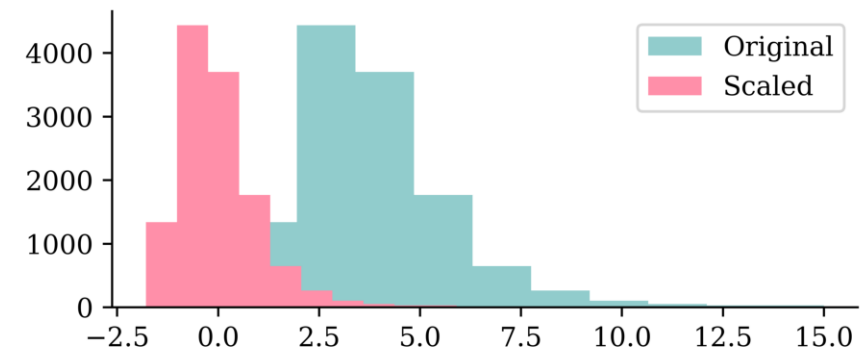
# Lecture Outline

- Artificial Intelligence
- Deep Learning Successes
- Types of Machine Learning Tasks
- Neural Networks
- California House Price Prediction
- Our First Neural Network
- Force positive predictions
- **Preprocessing**
- Early Stopping

# Re-scaling the inputs

```
1 scaler = StandardScaler()
2 scaler.fit(X_train)
3
4 X_train_sc = scaler.transform(X_train)
5 X_val_sc = scaler.transform(X_val)
6 X_test_sc = scaler.transform(X_test)
```

```
1 plt.hist(X_train.iloc[:, 0])
2 plt.hist(X_train_sc[:, 0])
3 plt.legend(["Original", "Scaled"]);
```



# Same model with scaled inputs

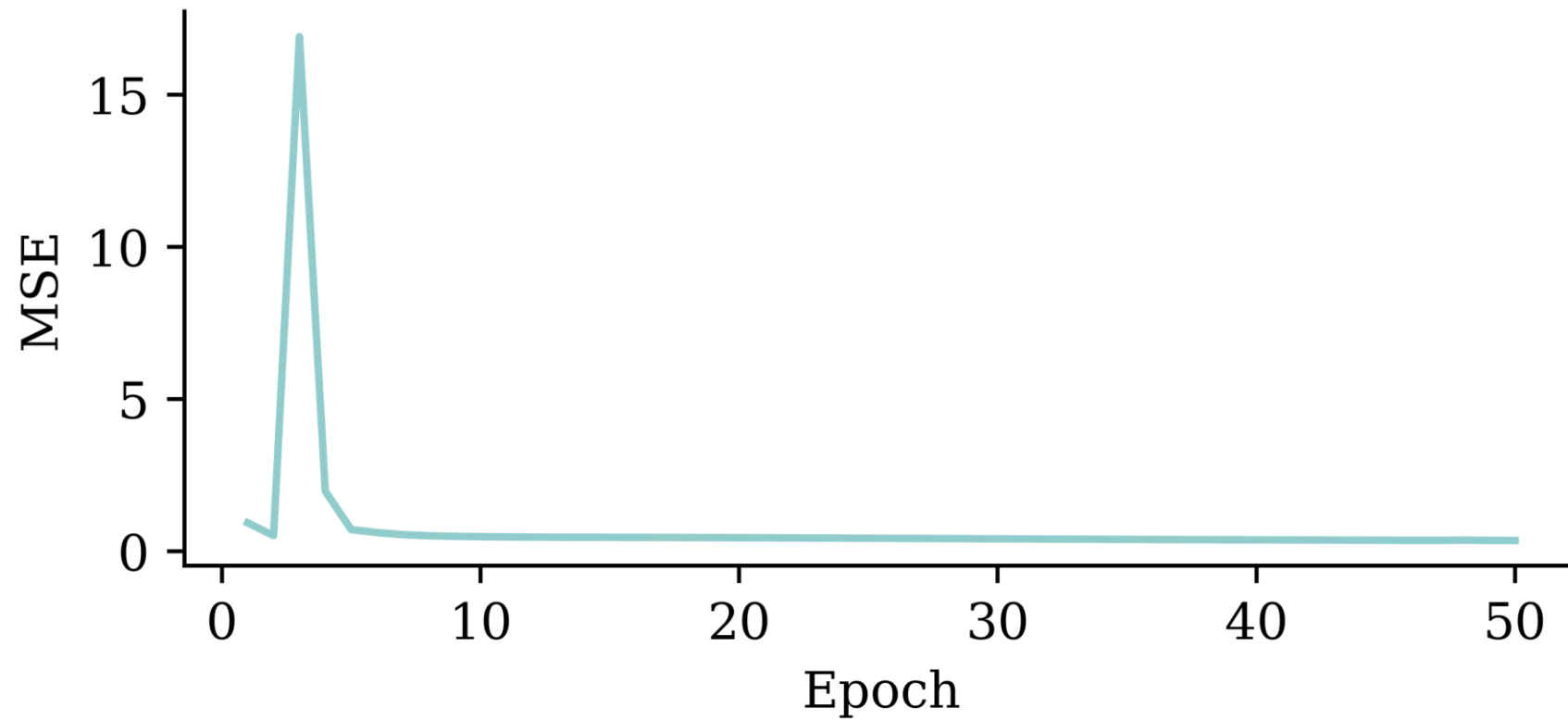
```
1 random.seed(2026)
2
3 model = Sequential([
4     Dense(30, activation="leaky_relu"),
5     Dense(1, activation="exponential")
6 ])
7
8 model.compile("adam", "mse")
9
10 %time hist = model.fit(X_train_sc, y_train, epochs=50, verbose=0)
```

CPU times: user 33.1 s, sys: 718 ms, total: 33.8 s  
Wall time: 33.2 s

Note the use of `X_train_sc` instead of `X_train`.

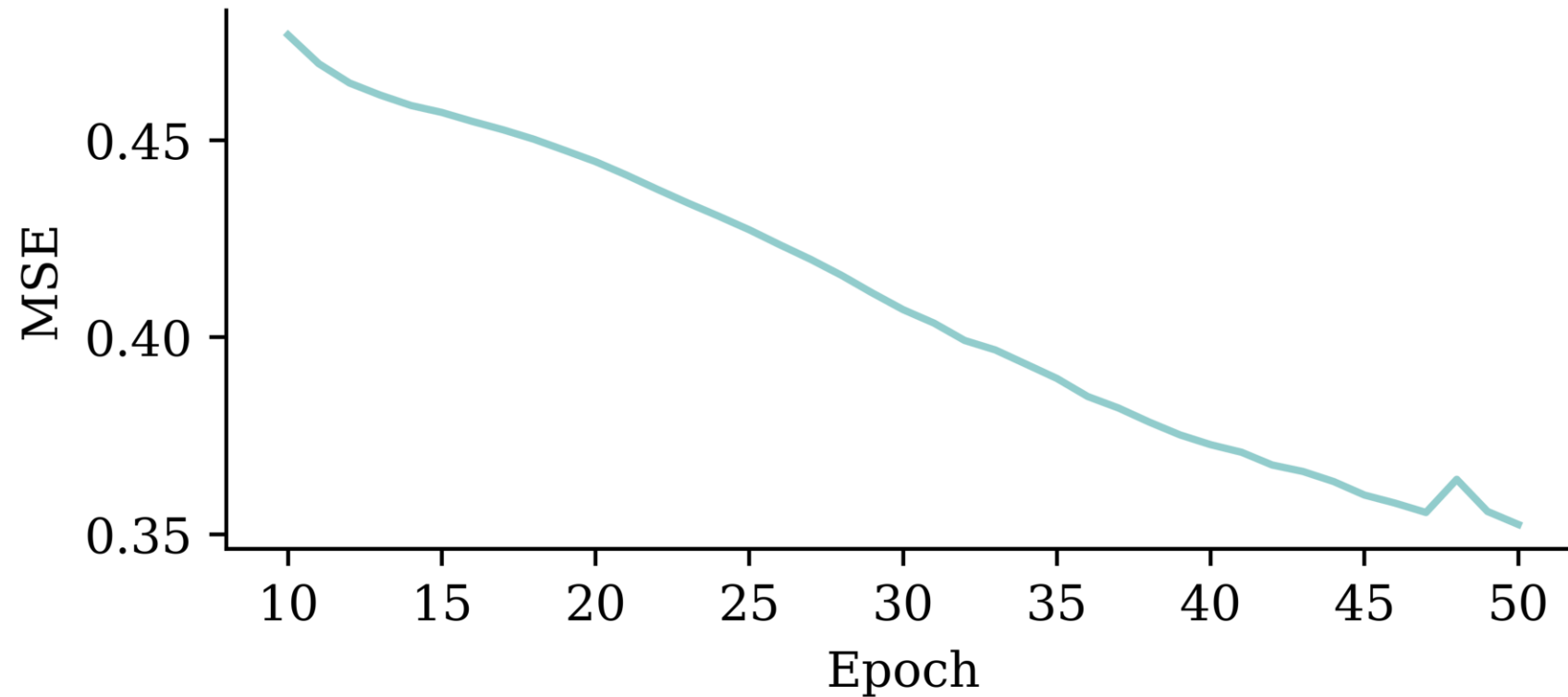
# Loss curve

```
1 plt.plot(range(1, 51), hist.history["loss"])
2 plt.xlabel("Epoch")
3 plt.ylabel("MSE");
```



# Loss curve

```
1 plt.plot(range(10, 51), hist.history["loss"][9:])  
2 plt.xlabel("Epoch")  
3 plt.ylabel("MSE");
```



# Predictions

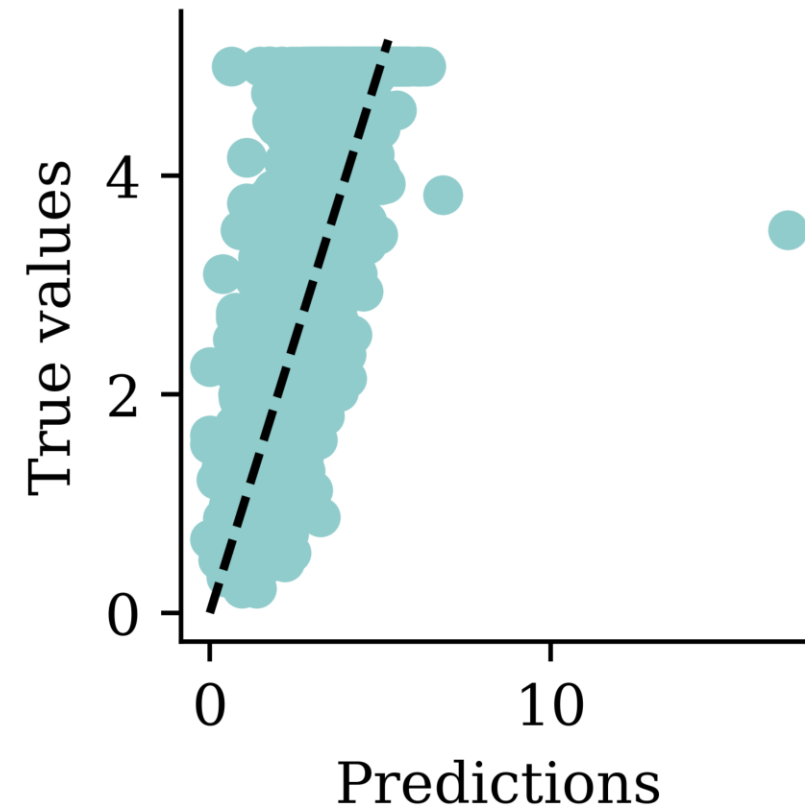
```
1 y_pred = model.predict(X_val_sc, verbose=0)
2 print(f"Min prediction: {y_pred.min():.2f}")
3 print(f"Max prediction: {y_pred.max():.2f}")
```

Min prediction: 0.00

Max prediction: 16.96

```
1 plt.scatter(y_pred, y_val)
2 plt.xlabel("Predictions")
3 plt.ylabel("True values")
4 add_diagonal_line()
```

```
1 mse_train["Exp ANN"] = mean_squared_error(
2     y_train, model.predict(X_train_sc, ve
3 )
4 mse_val["Exp ANN"] = mean_squared_error(y
```





# Comparing MSE (smaller is better)

On training data:

```
1 mse_train
```

```
{'Linear Regression': 0.5291948207479792,  
 'Basic ANN': 1.3506622360084013,  
 'Long run ANN': 0.6312395755307988,  
 'Exp ANN': 0.34680377119156724}
```

On validation data (expect *worse*, i.e. bigger):

```
1 mse_val
```

```
{'Linear Regression': 0.5059420205381369,  
 'Basic ANN': 1.322503585825187,  
 'Long run ANN': 0.629845824224333,  
 'Exp ANN': 0.38079265031009163}
```

# Comparing models (train)

```
1 train_results = pd.DataFrame({"Model": mse_train.keys(), "MSE": mse_train.values()})  
2 train_results.sort_values("MSE", ascending=False)
```

|   | Model             | MSE      |
|---|-------------------|----------|
| 1 | Basic ANN         | 1.350662 |
| 2 | Long run ANN      | 0.631240 |
| 0 | Linear Regression | 0.529195 |
| 3 | Exp ANN           | 0.346804 |

# Comparing models (validation)

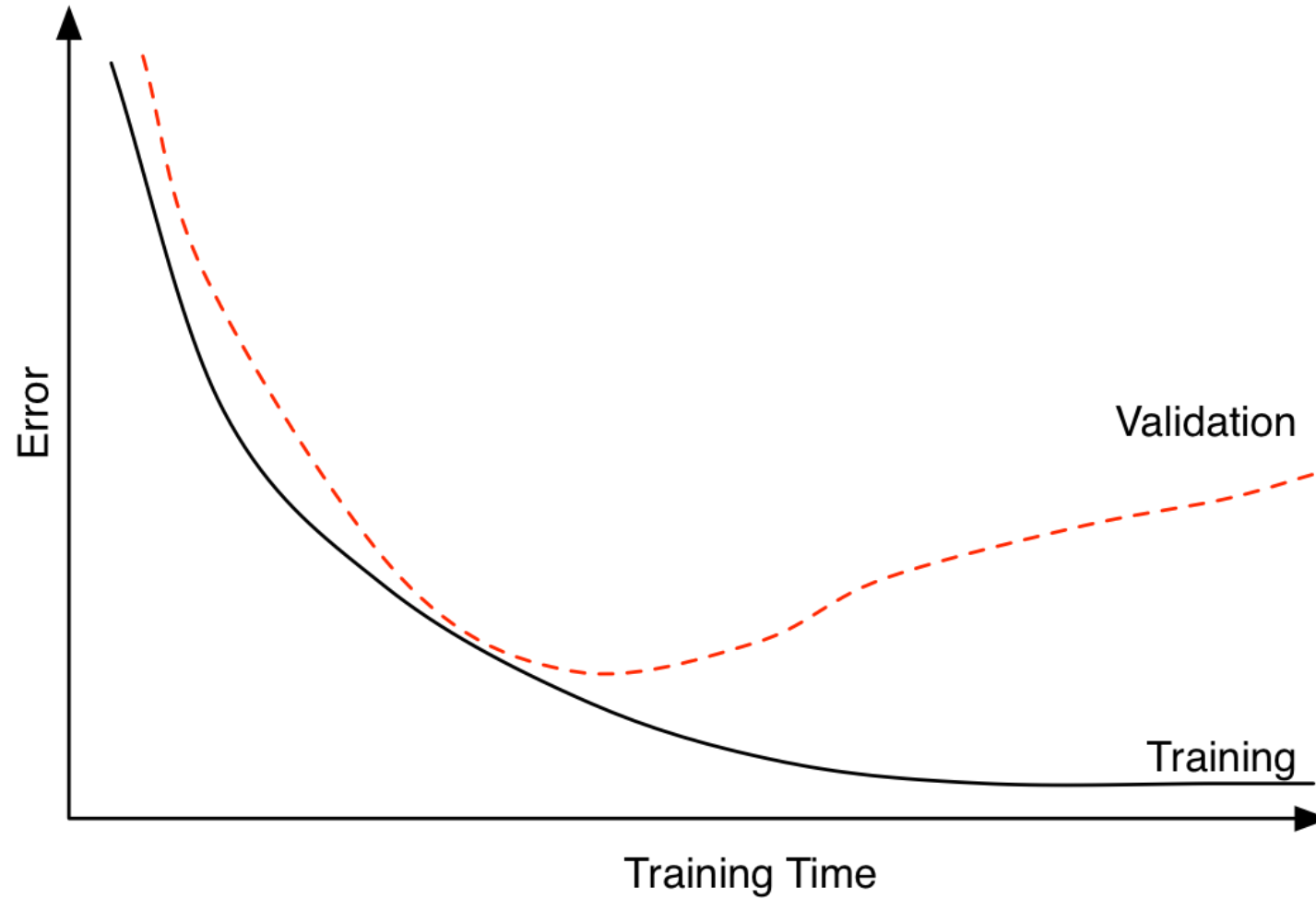
```
1 val_results = pd.DataFrame({"Model": mse_val.keys(), "MSE": mse_val.values()})
2 val_results.sort_values("MSE", ascending=False)
```

|   | Model             | MSE      |
|---|-------------------|----------|
| 1 | Basic ANN         | 1.322504 |
| 2 | Long run ANN      | 0.629846 |
| 0 | Linear Regression | 0.505942 |
| 3 | Exp ANN           | 0.380793 |

# Lecture Outline

- Artificial Intelligence
- Deep Learning Successes
- Types of Machine Learning Tasks
- Neural Networks
- California House Price Prediction
- Our First Neural Network
- Force positive predictions
- Preprocessing
- **Early Stopping**

# Choosing when to stop training



Illustrative loss curves over time.

Source: Heaton (2022), [Applications of Deep Learning](#), Part 3.4: Early Stopping.

# Try early stopping

Hinton calls it a “beautiful free lunch”

```
1 random.seed(2026)
2 model = Sequential([
3     Dense(30, activation="leaky_relu"),
4     Dense(1, activation="exponential")
5 ])
6 model.compile("adam", "mse")
7
8 es = EarlyStopping(restore_best_weights=True, patience=15)
9
10 %time hist = model.fit(X_train_sc, y_train, epochs=1_000, \
11     callbacks=[es], validation_data=(X_val_sc, y_val), verbose=0)
12 print(f"Keeping model at epoch #{len(hist.history['loss'])-15}.")
```

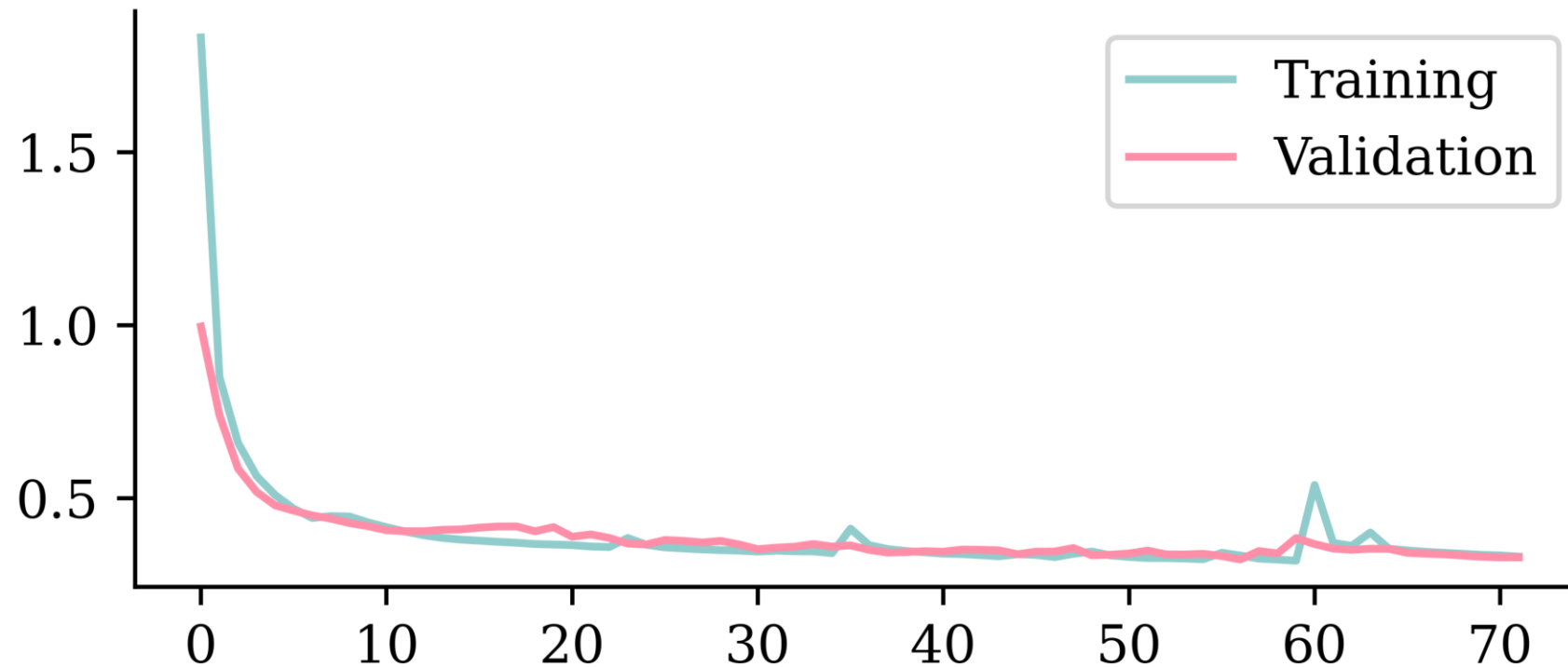
CPU times: user 1min, sys: 1.37 s, total: 1min 1s

Wall time: 1min

Keeping model at epoch #57.

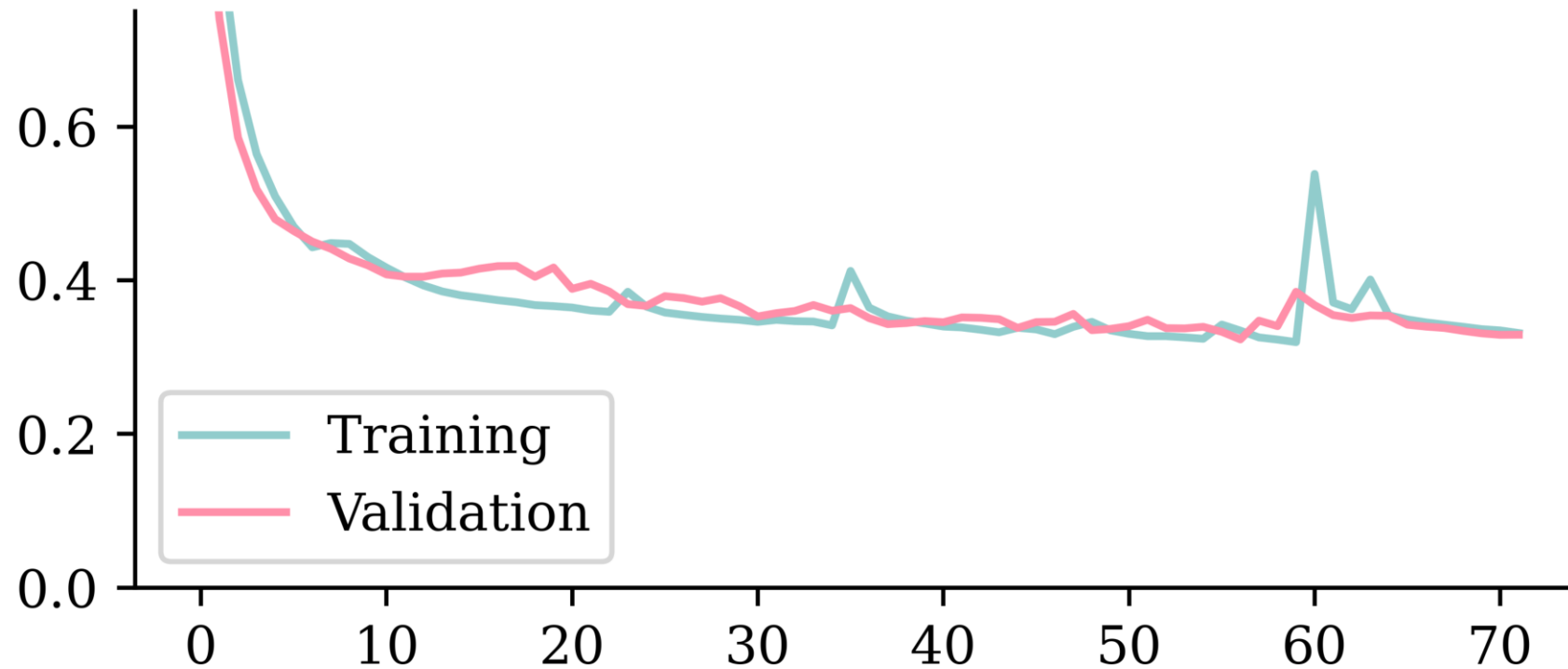
# Loss curve

```
1 plt.plot(hist.history["loss"])
2 plt.plot(hist.history["val_loss"])
3 plt.legend(["Training", "Validation"]);
```



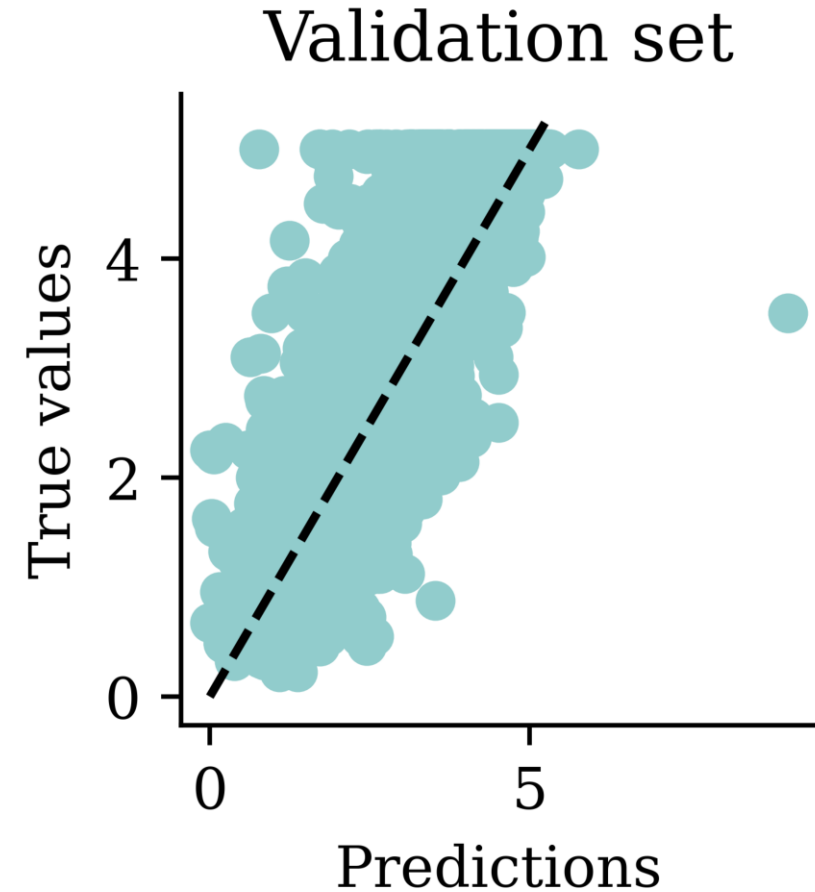
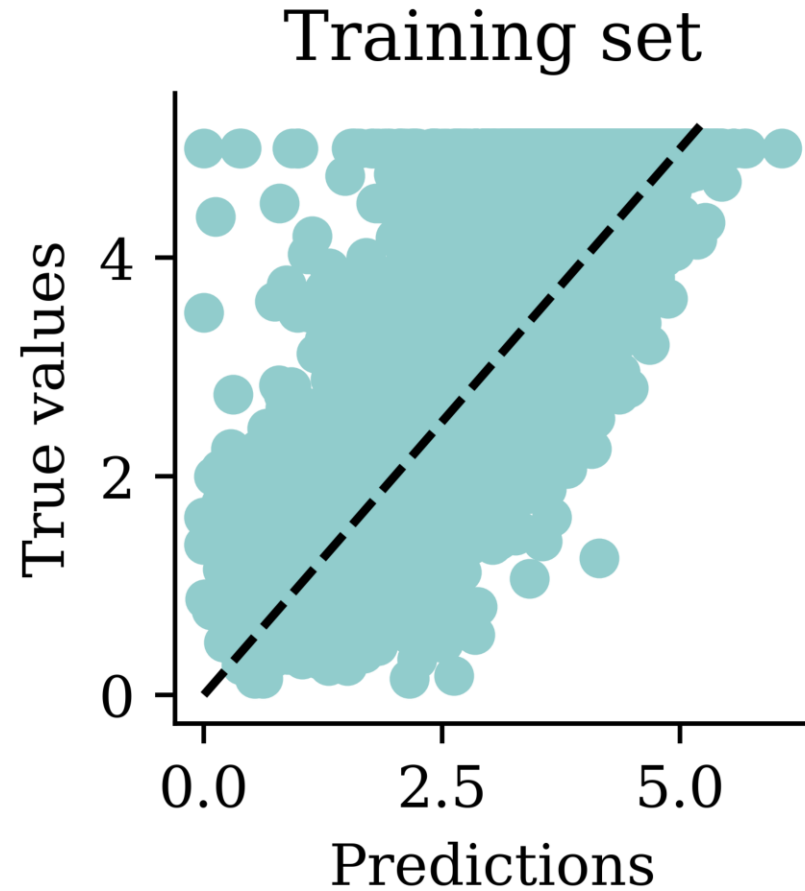
# Loss curve II

```
1 plt.plot(hist.history["loss"])
2 plt.plot(hist.history["val_loss"])
3 plt.ylim([0, 0.75])
4 plt.legend(["Training", "Validation"]);
```





# Predictions



# Comparing models (validation)

|          | Model             | MSE      |
|----------|-------------------|----------|
| <b>1</b> | Basic ANN         | 1.322504 |
| <b>2</b> | Long run ANN      | 0.629846 |
| <b>0</b> | Linear Regression | 0.505942 |
| <b>3</b> | Exp ANN           | 0.380793 |
| <b>4</b> | Early stop ANN    | 0.323157 |

# The test set

Evaluate *only the final/selected model* on the test set.

```
1 mean_squared_error(y_test, model.predict(X_test_sc, verbose=0))
```

```
0.33164115325909604
```

```
1 model.evaluate(X_test_sc, y_test, verbose=0)
```

```
0.3316410779953003
```

# Keras model methods

- `compile`: specify the loss function and optimiser
- `fit`: learn the parameters of the model
- `predict`: apply the model
- `evaluate`: apply the model and calculate a metric

```
1 random.seed(12)
2 model = Sequential()
3 model.add(Dense(1, activation="relu"))
4 model.compile("adam", "poisson")
5 model.fit(X_train, y_train, verbose=0)
6 y_pred = model.predict(X_val, verbose=0)
7 print(model.evaluate(X_val, y_val, verbose=0))
```

4.4610700607299805

# Package Versions

```
1 from watermark import watermark
2 print(watermark(python=True, packages="keras,matplotlib,numpy,pandas,seaborn,scipy,torch")
```

Python implementation: CPython

Python version : 3.14.3

IPython version : 9.13.0

keras : 3.14.1

matplotlib: 3.10.9

numpy : 2.4.4

pandas : 3.0.2

seaborn : 0.13.2

scipy : 1.17.1

torch : 2.11.0

# Glossary

- activations, activation function
- artificial neural network
- biases (in neurons)
- callbacks
- classification problem
- cost/loss function
- deep network, network depth
- dense or fully-connected layer
- early stopping
- epoch
- feed-forward neural network
- hidden layer
- Keras, TensorFlow, PyTorch
- labelled/unlabelled data
- machine learning
- minimax algorithm
- neural network architecture
- perceptron
- ReLU
- representation learning
- sigmoid activation function
- targets
- training/validation/test split
- weights (in a neuron)

# References

- Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., & Bengio, Y. (2014). Generative adversarial nets. *Advances in Neural Information Processing Systems*, 27.
- Ho, J., Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, 33, 6840–6851.
- Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). ImageNet classification with deep convolutional neural networks. *Advances in Neural Information Processing Systems*, 25, 1097–1105.
- Murphy, K. P. (2012). *Machine learning: A probabilistic perspective*. MIT Press.
- Russell, S., & Norvig, P. (2021). *Artificial intelligence: A modern approach* (4th ed.). Pearson.
- Samuel, A. L. (1959). Some studies in machine learning using the game of checkers. *IBM Journal of Research and Development*, 3(3), 210–229.